THE FOURIER ENTROPY–INFLUENCE INEQUALITY FOR READ-k DNF FORMULAS AND REGULAR BOOLEAN FUNCTIONS

YITZCHAK SHMALO

Abstract. The Fourier entropy–influence conjecture of Friedgut and Kalai asserts that the spectral entropy of a Boolean function is at most a universal constant times its total influence. We prove it with constant 15.1 for regular Boolean functions, a class that contains all linear threshold functions, and with constant $O(\log k)$ for read-k DNF formulas and, more generally, for unions of regular functions in which every variable is relevant to at most k of them; for Boolean functions of m regular functions the constant is $O(\log m)$. These classes satisfy the stronger inequality of O'Donnell and Tan, so by their composition theorem the conjecture also holds for read-once formulas built from such functions and gates of bounded arity, including iterated majorities of any odd fan-in. Threshold gates of any fan-in satisfy the biased form of that inequality under every product measure, so the conjecture also holds, with an absolute constant, for read-once formulas of any depth over threshold gates of unbounded fan-in, including iterated threshold functions with arbitrary thresholds and fan-ins.

1. Introduction

For $f : \{0,1\}^n \to \{0,1\}$ put $g = 1 - 2f \in \{-1,1\}$ and expand $g = \sum_{A\subseteq[n]} \widehat{g}(A)\chi_A$ with $\chi_A(x) = (-1)^{\sum_{i\in A} x_i}$. By Parseval the numbers $P(A) = \widehat{g}(A)^2$ form a probability distribution on subsets of $[n]$, the *spectral distribution* of f. Its Shannon entropy and its mean size are

$$H(f) = \sum_A P(A)\log_2 \frac{1}{P(A)}, \qquad I(f) = \sum_A |A|\,P(A) = \sum_{i=1}^{n} \mathrm{Inf}_i(f),$$

where $\mathrm{Inf}_i(f) = \mathbb{P}[f(x) \neq f(x \oplus e_i)]$ for uniform x. Friedgut and Kalai [6] conjectured that $H(f) \le C\,I(f)$ for a universal constant C (the FEI conjecture). The conjecture is open. It is known for symmetric functions and read-once decision trees [22], read-once formulas over gates of bounded arity [21] and over AND, OR, XOR and NOT gates [4], read-k decision trees [26, 24], random DNF formulas and random linear threshold functions [15, 3], plateaued functions [23], and δ-tribes functions and functions with separation properties [7]. For every Boolean function, Kelman, Kindler, Lifshitz, Minzer and Safra [14] proved $H(f) \le K\sum_A |A|\log_2(|A|+1)P(A) + K\,I(f)$ with an absolute constant K, which misses the conjecture by a factor logarithmic in the level. The subadditivity of entropy gives $H(f) \le \sum_i h(\mathrm{Inf}_i(f)) = O\big(I(f) + \sum_i \mathrm{Inf}_i(f)\log(1/\mathrm{Inf}_i(f))\big)$, with h the binary entropy [16]; Han [8] proved the second bound by an iteration over the levels. The best known lower bound on C is 6.5218 [20], following 6.278 [21] and 6.4548 [12].

The conjecture implies Mansour's conjecture on the Fourier spectrum of DNF formulas [17, 22]. Mansour's conjecture is known for read-k DNF formulas [15]. The weaker Fourier min-entropy–influence inequality, $\log_2(1/\max_A P(A)) \le C\,I(f)$, was proved for regular read-k DNF formulas by Shalev [24] and for all read-k DNF formulas, with $C = O(\log k)$, by Arunachalam, Chakraborty, Koucký, Saurabh and de Wolf [1]. The entropy inequality itself was open for this class; Shalev [24] proposed it as a direction.

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A DNF formula is a disjunction $\varphi = T_1 \vee \cdots \vee T_m$ of terms, each term T_t being the conjunction of literals on a nonempty set V_t of variables. It is *read-k* if every variable lies in at most k of the sets V_t. No minimality is assumed.

Theorem 1.1. *Let f be computed by a read-k DNF formula. Then*

$$H(f) \le \min\bigl(2.89 + 9.9\,k,\ 30 + 5.6 \log k\bigr)\, I(f).$$

The same bound holds for functions computed by read-k CNF formulas.

The dependence on k is logarithmic, as in the min-entropy bound of [1]. A DNF formula with m terms is read-m, so every such formula satisfies $H(f) \le (30 + 5.6 \log m)\, I(f)$.

For linear threshold functions the conjecture was open: Chakraborty, Kulkarni, Lokam and Saurabh [4] proved $H(f) = O(\sqrt{n})$ for them, and [3] treated random ones. A monotone Boolean function f is *regular* (or 2-monotone [19]) if its coordinates are totally preordered by desirability: for all $i \ne j$ either $f(x^{i\leftarrow 1, j\leftarrow 0}) \ge f(x^{i\leftarrow 0, j\leftarrow 1})$ for every x, or the reverse inequality holds for every x. A function is regular if it becomes a monotone regular function after negating some of its inputs. Every linear threshold function is regular.

Theorem 1.2. *Every regular Boolean function satisfies $H(f) \le 15.1\, I(f)$.*

The proof extends to Boolean combinations of regular functions whose desirability orders are unrelated (Theorem 3.7): if $h_1, \dots, h_m$ are regular and $F : \{0,1\}^m \to \{0,1\}$ is arbitrary, then $f = F(h_1, \dots, h_m)$ satisfies $H(f) \le (16.5 + 8.4 \ln m)\, I(f)$. In particular the conjecture holds, with constant $O(\log m)$, for intersections of m halfspaces. For unions and intersections the dependence on m can be replaced by the overlap: if $h_1, \dots, h_m$ are regular and every variable is relevant to at most k of them, then $h_1 \vee \cdots \vee h_m$ and $h_1 \wedge \cdots \wedge h_m$ satisfy $H(f) \le \min(3.9{+}21.2k,\ 54{+}12.8 \log k)\, I(f)$ (Theorems 4.1 and 4.4). This contains Theorems 1.1 and 1.2 up to the constants. The two statements combine: if every h_l is a Boolean function of at most M regular functions, the constant is $54 + 12.8 \log k + 6.8 \log M$ (Theorem 4.5); for example, for unions of polytopes with M facets in which every variable enters at most k of them, and, since every function of w variables is a function of w dictators, for read-k unions of arbitrary functions of at most w variables each.

O'Donnell and Tan [21] introduced a stronger inequality that is preserved under disjoint composition. For the uniform measure it reads

$$H^+(f) := \sum_{A \ne \emptyset} P(A) \log_2 \frac{1}{P(A)} \le C\,\bigl(I(f) - \mathrm{Var}(f)\bigr), \tag{1}$$

where $\mathrm{Var}(f) = 1 - P(\emptyset)$, and it implies $H(f) \le \max(C, 1/\ln 2)\, I(f)$ [21, Proposition 2.1].

Theorem 1.3. *Inequality (1) holds with $C = \min(22 + 28k,\ 88 + 16.4 \log k)$ for functions computed by read-k DNF or CNF formulas, with $C = 60$ for regular functions, with $C = \min(24 + 66k,\ 186 + 46.2 \log k)$ for unions and intersections of regular functions in which every variable is relevant to at most k of them, and with $C = 66 + 31 \ln m$ for Boolean functions of m regular functions.*

Together with the composition theorem of [21], this gives the conjecture for a class that contains read-once formulas, read-k DNF formulas and intersections of halfspaces.

Corollary 1.4. *Let F be a read-once formula and $f = F(f_1, \dots, f_s)$, where $f_1, \dots, f_s$ depend on pairwise disjoint sets of variables and each f_l belongs to one of the classes of Theorem 1.3. Suppose that every gate of F is an* AND*,* OR *or* XOR *gate of any fan-in, or has arity at most a, or is a threshold gate $T_{t,r}(y) = \mathbf{1}[y_1 + \cdots + y_r \ge t]$ of any fan-in r, or belongs to one of these classes and has only balanced inputs, that is, inputs equal to 1 with probability $\frac{1}{2}$. Then*

$H(f) \le C\, I(f)$, where C is the largest of $C_a = 2^{O(a)}$, of 760 *if threshold gates occur (*66.4 *if the inputs of each threshold gate are all equal to* 1 *with the same probability), and of the constants of Theorem 1.3 for the classes that occur.*

For instance, the corollary applies to unions and intersections of any number of halfspaces on pairwise disjoint sets of variables, to majorities of intersections of halfspaces on disjoint sets of variables, and to iterated majorities $\mathrm{MAJ}_{r_1} \circ \mathrm{MAJ}_{r_2} \circ \cdots \circ \mathrm{MAJ}_{r_d}$ with odd fan-ins of any size, since a majority of an odd number of independent balanced bits is balanced; in these examples the constant is 66, independent of the number of halfspaces and of the fan-ins. Threshold gates satisfy the biased form of (1) under every product measure, with constant 66.4 when their inputs have a common bias (Theorem 5.6) and with constant 760 in general (Theorem 5.13). So the corollary also applies to iterated threshold functions $T_{t_1,r_1} \circ T_{t_2,r_2} \circ \cdots \circ T_{t_d,r_d}$ with arbitrary thresholds and fan-ins, where the bias changes from level to level, and to thresholds of conjunctions of equal width, with constant 66.4, and to every read-once formula over threshold gates of any fan-in, with constant 760 (Corollary 5.14). The constant 760 comes from splitting at pivotal probability $\frac{9}{10}$: above it the other inputs of the gate are nearly deterministic and a direct estimate applies, and below it the weak form loses a factor 10. The weaker inequality of Proposition 5.10, with I_μ in place of $I_\mu - \mathrm{Var}_\mu$, composes at the cost of adding its constant at each level (Lemma 5.11) and gives $H(f) \le (38d + \log e)\, I(f)$ for formulas of depth d (Corollary 5.12), which is smaller when $d \le 19$. Gates AND, OR and XOR of any fan-in are read-once formulas over gates of arity 2, so [21] already covers them with an unspecified constant; Lemma 5.5 gives them the biased form of (1) with the sharp constant $2 + 4\log e$, which makes the constant of the corollary explicit for formulas over these gates.

The proofs of the two theorems use different mechanisms. For Theorem 1.1, the discrete derivative $D_i g$ is supported inside the union of at most k subcubes, one for each term containing i, and on each subcube it factorises exactly into the indicator of the subcube times a function of the remaining coordinates. Consequently the spectral sample, conditioned on containing i, contains a uniformly random subset of the variables of some term through i (Lemma 2.2). This is combined with a Gibbs inequality against a reference measure that activates terms independently, and with a path argument that bounds the cost of the terms that are not activated (Lemma 2.1). For the logarithmic bound only one term of each connected piece of the cover is activated; the other terms of the piece are reached through shared variables, at a cost of order $\log k$ each (Theorem 2.5). For Theorem 1.2, the pivotal set of a coordinate, restricted to the coordinates that are at least as desirable, is an antichain; Sperner's theorem then shows that the spectral sample, conditioned on containing i, contains a positive proportion of the more desirable coordinates in a logarithmic average (Lemma 3.4), and a rank-weighted reference measure converts this into an entropy bound. Theorem 4.1 combines the two: the reference measure activates the functions h_l, and the uniform filling of a witnessing term is replaced by a level inequality for functions supported on the pivotal set of a rarely true h_l (Lemma 4.3); Theorem 4.4 adds the rooted activation of Theorem 2.5 and rank laws normalised at the size of the part of S assigned to each h_l. For Theorem 1.3, two lemmas valid for every Boolean function reduce (1) to the sets with at least two elements (Lemmas 5.1 and 5.2); on those sets both proofs go through with small changes (Section 5).

Notation and standard facts. Throughout, x is uniform on $\{0,1\}^n$, $\log = \log_2$, and ln is the natural logarithm. S denotes a random set with law P (the spectral sample of f), so $H(f) = H(S)$, $I(f) = \mathbb{E}|S|$ and $\mathbb{P}[i \in S] = \mathrm{Inf}_i(f)$. We write $\mathrm{Var}(f) = 1 - P(\emptyset) = 4p(1-p)$ with $p = \mathbb{P}[f = 1]$, and h for the binary entropy function in bits.

For $i \in [n]$ the discrete derivative

$$D_i g(x) = \tfrac{1}{2}\big(g(x^{i\leftarrow 0}) - g(x^{i\leftarrow 1})\big) = f(x^{i\leftarrow 1}) - f(x^{i\leftarrow 0})$$

does not depend on x_i, takes values in $\{-1, 0, 1\}$, and is nonzero exactly on the set $\mathcal{P}_i$ of points at which i is pivotal. Since $g = \mathbb{E}_i g + \chi_{\{i\}} D_i g$, we have $\widehat{g}(A) = \widehat{D_i g}(A \setminus \{i\})$ for $A \ni i$. Hence $\mathrm{Inf}_i(f) = \mathbb{E}[(D_i g)^2] = \mathbb{P}[\mathcal{P}_i]$, and the law ν_i of $S \setminus \{i\}$ given $i \in S$ is $\nu_i(T) = \widehat{D_i g}(T)^2/\mathrm{Inf}_i(f)$.

Gibbs' inequality. If $Q \geq 0$ is a function on subsets of $[n]$ with $\sum_A Q(A) \leq 1$ and $Q(A) > 0$ whenever $P(A) > 0$, then $H(f) \leq \sum_A P(A) \log(1/Q(A))$.

Edge isoperimetry [9, 2, 10]. A set $B \subseteq \{0,1\}^n$ of density α has at least $\alpha 2^n \log(1/\alpha)$ edges of the cube leaving it. Since $I(f) = 2^{1-n} \cdot \#\{\text{edges between } f^{-1}(0) \text{ and } f^{-1}(1)\}$, we get $I(f) \geq 2\alpha \log(1/\alpha)$ with $\alpha = \min(p, 1-p)$.

Lemma 1.5. *$h(\mathrm{Var}(f)) \leq 2.8854\, I(f)$ and $H(|S|) \leq 3.8854\, I(f)$.*

Proof. Let $\alpha = \min(p, 1-p) \leq 1/2$ and $\varepsilon = \mathrm{Var}(f) = 4\alpha(1-\alpha) \in [2\alpha, 4\alpha]$. Using $(1-\varepsilon)\ln\frac{1}{1-\varepsilon} \leq \varepsilon$,

$$h(\varepsilon) \leq \varepsilon \log\frac{1}{\varepsilon} + \varepsilon \log e \leq 4\alpha\Big(\log\frac{1}{\alpha} - 1\Big) + 4\alpha \log e \leq 2I(f) + 1.7708\,\alpha \leq 2.8854\, I(f),$$

where we used $I(f) \geq 2\alpha\log(1/\alpha)$ and $I(f) \geq \mathrm{Var}(f) \geq 2\alpha$. For the second bound, the chain rule gives $H(|S|) = h(\varepsilon) + \varepsilon\, H(|S| \mid S \neq \emptyset)$, and for a random variable K with values in $\{1, 2, \dots\}$ Gibbs' inequality against the law 2^{-k} gives $H(K) \leq \mathbb{E}K$. Hence $\varepsilon\, H(|S| \mid S \neq \emptyset) \leq \varepsilon\, \mathbb{E}[|S| \mid S \neq \emptyset] = I(f)$. □

2. Read-k DNF formulas

Fix a read-k DNF formula computing f. Term t is satisfied exactly when x agrees with a fixed assignment $a_t \in \{0,1\}^{V_t}$ on V_t. Put $p_0 = \mathbb{P}[f = 0]$ and $\mu = \sum_t 2^{-|V_t|}$.

Lemma 2.1 (Path lemma)**.** *$p_0\, \mu \leq \frac{k}{4}\, I(f)$.*

Proof. Fix an order on each V_t. For a pair (x, t) with $f(x) = 0$, change the coordinates of V_t at which x differs from a_t one at a time, in this order. The function f is 0 at the start and 1 at the end, so the path contains a first edge $\{y, y \oplus e_j\}$ with $f(y) = 0$ and $f(y \oplus e_j) = 1$, and $j \in V_t$. Given this edge and t, the point x agrees with y outside the set of coordinates of V_t that precede j, so at most $2^{|V_t|-1}$ pairs (x, t) are mapped to the same edge and the same t. Since at most k terms contain j,

$$\sum_{x:\, f(x)=0} \sum_t 2^{-|V_t|} \leq \sum_{\text{boundary edges}} \sum_{t:\, j \in V_t} 2^{|V_t|-1}\, 2^{-|V_t|} \leq \frac{k}{2} \cdot 2^{n-1} I(f).$$

Dividing by 2^n proves the lemma. □

Lemma 2.2 (Witness decomposition)**.** *Let $i \in [n]$ with $\mathrm{Inf}_i(f) > 0$, and let $t_1, \dots, t_r$ ($r \leq k$) be the terms whose variable sets contain i. For $\beta > 1$ let E_i be the set of $T \subseteq [n] \setminus \{i\}$ such that $|(T \cup \{i\}) \cap V_{t_s}| < |V_{t_s}|/\beta$ for every s. Then*

$$\nu_i(E_i) \leq \sum_{s=1}^{r} \mathbb{P}\Big[\mathrm{Bin}\big(|V_{t_s}| - 1, \tfrac{1}{2}\big) < \tfrac{|V_{t_s}|}{\beta} - 1\Big].$$

Proof. If i is pivotal at x, some term is satisfied at exactly one of $x^{i\leftarrow 0}, x^{i\leftarrow 1}$. That term depends on x_i, so it is some t_s, and x lies in the subcube $C_s = \{x : x_j = a_{t_s}(j) \text{ for } j \in W_s\}$, where $W_s = V_{t_s} \setminus \{i\}$. Let $\mathcal{P}_{i,s}$ be the set of pivotal points whose least such index is s, and put

$d_s = D_i g \cdot 1_{\mathcal{P}_{i,s}}$. The sets $\mathcal{P}_{i,s}$ partition $\mathcal{P}_i$, so $D_i g = \sum_s d_s$ and $\mathrm{Inf}_i(f) = \sum_s \lambda_s \mathrm{Inf}_i(f)$ with $\lambda_s = \|d_s\|_2^2/\mathrm{Inf}_i(f)$.

Write $x = (y, z)$ with y the restriction of x to W_s. Because $\mathcal{P}_{i,s} \subseteq C_s$,

$$d_s(y, z) = \mathbf{1}[y = a_{t_s}|_{W_s}] \cdot \psi_s(z), \qquad \psi_s(z) = d_s(a_{t_s}|_{W_s}, z).$$

The two factors depend on disjoint sets of coordinates, so Fourier coefficients multiply, and every Fourier coefficient of the indicator of a point of $\{0,1\}^{W_s}$ has absolute value $2^{-|W_s|}$. Therefore $\widehat{d_s}(T)^2 = 4^{-|W_s|}\,\widehat{\psi_s}(T \setminus W_s)^2$: under the probability measure $\nu_{i,s}(T) = \widehat{d_s}(T)^2/\|d_s\|_2^2$ the set $T \cap W_s$ is uniformly distributed over the subsets of W_s. By the Cauchy–Schwarz inequality, for every T, $\big(\sum_s \widehat{d_s}(T)\big)^2 \le \big(\sum_s \lambda_s\big)\big(\sum_s \widehat{d_s}(T)^2/\lambda_s\big)$, the sum running over s with $\lambda_s > 0$. Summing over $T \in E_i$ gives $\nu_i(E_i) \le \sum_s \nu_{i,s}(E_i)$. Finally $E_i \subseteq \{T : |T \cap W_s| < |V_{t_s}|/\beta - 1\}$, and $|T \cap W_s|$ has the law $\mathrm{Bin}(|W_s|, 1/2)$ under $\nu_{i,s}$. □

For $\beta > 2$ put

$$C_\beta = \sup_{m \ge 1} m\,\mathbb{P}\Big[\mathrm{Bin}\big(m-1, \tfrac12\big) < \tfrac{m}{\beta} - 1\Big],$$

which is finite by the Chernoff bound. A direct computation gives $C_3 = 13 \cdot 299/4096 = 0.9489\ldots$, attained at $m = 13$.

Theorem 2.3. *Let f be computed by a read-k DNF formula and let $\beta > 2$. Then*

$$H(f) \le \big(2.886 + k\,(2 + 2\beta + 2C_\beta)\big)\, I(f).$$

With $\beta = 3$ this gives $H(f) \le (2.886 + 9.898\,k)\, I(f)$.

Proof. Reference measure. For a set U of terms let $V(U)$ be the union of their variable sets and $q(U) = \prod_{t \in U} \pi_t \prod_{t \notin U}(1 - \pi_t)$ with $\pi_t = 2^{-|V_t|}$. Put $\varepsilon = \mathrm{Var}(f)$, $Q(\emptyset) = 1 - \varepsilon$ and, for $A \neq \emptyset$, $Q(A) = \varepsilon\, Q_1(A)$ with

$$Q_1(A) = \sum_U q(U)\, 2^{-|V(U)|}\, \mathbf{1}[A \subseteq V(U)].$$

Summing first over $A \subseteq V(U)$ shows $\sum_A Q(A) \le 1$. A variable that lies in no term is irrelevant to f, so $Q(A) > 0$ whenever $P(A) > 0$. Gibbs' inequality gives

$$H(f) \le h(\varepsilon) + \sum_{A \neq \emptyset} P(A) \log \frac{1}{Q_1(A)}. \tag{2}$$

If U covers A, then $Q_1(A) \ge q(U) 2^{-|V(U)|}$, so

$$\log \frac{1}{Q_1(A)} \le \Lambda + \sum_{t \in U}\Big(\log \frac{1}{\pi_t} + |V_t|\Big) = \Lambda + 2\sum_{t \in U} |V_t|, \qquad \Lambda = \sum_t \log \frac{1}{1 - \pi_t} \le 2\mu, \tag{3}$$

where the last inequality holds because $\pi_t \le \frac12$ and $\log \frac{1}{1-x} \le 2x$ for $0 \le x \le \frac12$ by convexity.

The cover. For $A \neq \emptyset$ let $D(A)$ be the set of terms with $|A \cap V_t| \ge |V_t|/\beta$. Call $i \in A$ *sparse* if no term in $D(A)$ contains it, and let w_i be the smallest size of a term containing i. Cover A by $D(A)$ together with a smallest term through each sparse element. Since every variable lies in at most k terms, $\sum_{t \in D(A)} |V_t| \le \beta \sum_{t \in D(A)} |A \cap V_t| \le \beta k |A|$, and (3) gives

$$\log \frac{1}{Q_1(A)} \le \Lambda + 2\beta k |A| + 2 \sum_{i \in A \text{ sparse}} w_i.$$

Now $\sum_{A \ni i} P(A)\, \mathbf{1}[i \text{ sparse in } A] = \mathrm{Inf}_i(f)\, \nu_i(E_i)$ with E_i as in Lemma 2.2, and a sparse i lies only in terms of size larger than β. By Lemma 2.2 and $w_i \le |V_{t_s}|$,

$$w_i\, \nu_i(E_i) \le \sum_s |V_{t_s}| \mathbb{P}\Big[\mathrm{Bin}\big(|V_{t_s}| - 1, \tfrac12\big) < \tfrac{|V_{t_s}|}{\beta} - 1\Big] \le k\, C_\beta.$$

Summing over $A \neq \emptyset$,

$$\sum_{A \neq \emptyset} P(A) \log \frac{1}{Q_1(A)} \leq \varepsilon\Lambda + 2\beta k\, I(f) + 2kC_\beta\, I(f).$$

The vacuum term. By Lemma 2.1, $\varepsilon\Lambda \leq 4p_0 \cdot 2\mu \leq 2k\, I(f)$.

The empty set. $h(\varepsilon) \leq 2.886\, I(f)$ by Lemma 1.5. Adding the three bounds proves the theorem. □

Proof of Theorem 1.1. For DNF formulas the first bound is Theorem 2.3 with $\beta = 3$, and the second is Theorem 2.5, proved in Section 2.1. If f is computed by a read-k CNF formula, the dual function $f^*(x) = 1 - f(\bar{x})$, where $\bar{x}$ is the complement of x, is computed by a read-k DNF formula. Its ± 1 version is $x \mapsto -g(\bar{x})$, whose Fourier coefficients are $\pm\widehat{g}(A)$, so f and f^* have the same spectral distribution and the same total influence. □

Remark 2.4. (1) Averaging the proof of Lemma 2.1 over uniformly random orders of the sets V_t gives $p_0\mu \leq \frac{1}{2}\max_j\big(\sum_{t \ni j} |V_t|^{-1}\big) I(f)$, which improves the vacuum term when all terms are long. (2) For monotone formulas one can also take π_t to be the probability that t is the unique satisfied term divided by the probability that no other term is satisfied; Harris' inequality then bounds $\log(1/\pi_t)$ by $|V_t|$ plus the number of terms meeting V_t, which leads to a constant of order k^2. (3) The dependence on k enters in three places: the vacuum term, the Cauchy–Schwarz step, and the cover. For the star $x_c \wedge (x_1 \vee \cdots \vee x_k)$, written as the read-$k$ formula $\bigvee_j (x_c \wedge x_j)$, one has $p_0\mu \approx k/8$ while $I(f) \approx 1$, so Lemma 2.1 is sharp up to a factor 2, and the reference measure of the proof loses a factor of order k on this formula, which is a read-once formula.

2.1. **Logarithmic dependence on k.** The reference measure of Theorem 2.3 pays for every term of the cover twice, once to activate it and once for the variables it adds to $V(U)$, and the dense terms through a single variable can be as many as k. The following reference measure activates only one term in each connected piece of the cover, reaches the other terms of that piece through shared variables at a logarithmic price, and pays for the subset of $V(U)$ according to its density.

For $\beta > 2$ and integers $w \geq 1$ put $\tau_\beta(m) = \mathbb{P}[\mathrm{Bin}(m-1, \frac{1}{2}) < \frac{m}{\beta} - 1]$,

$$T_\beta(w) = \sup_{m \geq w} m\, \tau_\beta(m), \qquad M_\beta(k) = \sup_{w \geq 1} \min\big(w,\, k\, T_\beta(w)\big).$$

By the Chernoff bound $\tau_\beta(m) \leq e^{-(m-1)D_\beta}$, where $D_\beta = \frac{1}{\beta}\ln\frac{2}{\beta} + (1 - \frac{1}{\beta})\ln\big(2 - \frac{2}{\beta}\big)$ is the relative entropy of the Bernoulli laws with means $\frac{1}{\beta}$ and $\frac{1}{2}$. The function $m \mapsto me^{-(m-1)D_\beta}$ increases up to $1/D_\beta$ and decreases afterwards, so

$$M_\beta(k) \leq \max\Big(\frac{1}{D_\beta},\, 1 + \frac{\ln k}{D_\beta}\Big). \tag{4}$$

Indeed, if $w \geq w_0 = 1 + \ln k / D_\beta$ and $w_0 \geq 1/D_\beta$, then $k\, T_\beta(w) \leq kwe^{-(w-1)D_\beta} \leq w_0$, while $\min(w, kT_\beta(w)) \leq w$ always.

Theorem 2.5. *Let f be computed by a read-k DNF or CNF formula and let $\beta > 2$. Then*

$$H(f) \leq \Big(11.39 + \beta + \log k + \log(\beta k + 1) + \log(\beta k) + \kappa_\beta M_\beta(k)\Big)\, I(f),$$

where $\kappa_\beta = 1 + (\log e)\big(\frac{1}{\beta k} + \frac{1}{\beta k + 1}\big)$. In particular $H(f) \leq (30 + 5.6 \log k)\, I(f)$ for every $k \geq 1$.

Proof. By duality, as in the proof of Theorem 1.1, it suffices to treat DNF formulas. Put $\lambda = 1/k$, $\pi_t = 2^{-|V_t|}$ and $\varepsilon = \mathrm{Var}(f)$.

Reference measure. Consider the following random procedure, which may fail. (a) Choose a set R of terms, containing each term t independently with probability $\lambda\pi_t/(1+\lambda\pi_t)$. (b) Put the terms of R into a queue. When a term u is taken from the queue, choose an integer $c \ge 0$ with probability $1/((c+1)(c+2))$, a multiset of c variables of V_u uniformly among the $\binom{|V_u|+c-1}{c}$ such multisets, and for each chosen variable j a term containing j, uniformly among the at most k such terms; append the chosen terms to the queue. Let U be the set of terms taken from the queue once it is empty (the procedure fails if it never empties). (c) Choose $s \ge 1$ with probability $1/(s(s+1))$ and a uniformly random s-element subset of $V(U)$ (fail if $s > |V(U)|$). Let $Q_1(A)$ be the probability that the procedure outputs the set A; then $\sum_A Q_1(A) \le 1$. If a nonempty set A is the output of a particular run, $Q_1(A)$ is at least the probability of that run, and so

(5)
$$\log\frac{1}{Q_1(A)} \le \Lambda_\lambda + \sum_{r\in R}\log\frac{1}{\lambda\pi_r} + \sum_{u\in U}\Bigl(\log\bigl((c_u+1)(c_u+2)\bigr) + \log\binom{|V_u|+c_u-1}{c_u} + c_u\log k\Bigr) + \log\bigl(s(s+1)\bigr) + \log\binom{|V(U)|}{s},$$

where c_u is the integer chosen when u is processed and $\Lambda_\lambda = \sum_t \log(1+\lambda\pi_t) \le (\log e)\,\mu/k$.

A run producing A. Let $A \ne \emptyset$ with $P(A) > 0$, let $D(A)$ be the set of terms with $|A\cap V_t| \ge |V_t|/\beta$, and call $i \in A$ sparse if no term of $D(A)$ contains it, as in the proof of Theorem 2.3. Let $U_d \subseteq D(A)$ be a minimal family covering the non-sparse elements of A, and for each sparse i let t_i be a smallest term containing i, of size w_i. Put $U = U_d \cup \{t_i\}$. By minimality every term of U_d contains a non-sparse element of A that no other term of U_d contains, and no t_i belongs to $D(A)$; hence $|U| \le |A|$. In the graph on U in which two terms are adjacent when their variable sets meet, choose in each connected component a root and a breadth-first spanning tree, and let R be the set of roots. Processing the terms along these trees, each non-root term being chosen from its parent through a common variable, and then choosing $s = |A|$ and the subset A of $V(U)$, is a run that outputs A; in it c_u is the number of children of u, so $\sum_u c_u = |U| - |R|$.

Cost of the run. Put $s = |A|$ and $W = \sum_{i\in A \text{ sparse}} w_i$. The roots lie in different components, so their variable sets are pairwise disjoint; a root in U_d has $|V_r| \le \beta|A\cap V_r|$, and every other root is some t_i. Hence

$$\sum_{r\in R}\log\frac{1}{\lambda\pi_r} = \sum_{r\in R}\bigl(\log k + |V_r|\bigr) \le |R|\log k + \beta s + W.$$

Since $(c+1)(c+2) \le 2\cdot 3^c$, the first sum over u in (5) contributes at most $|U| + (|U|-|R|)\log 3 \le s(1+\log 3)$, and the terms $c_u \log k$ together with $|R|\log k$ give $|U|\log k \le s\log k$. Every variable lies in at most k terms, so $V_{\text{tot}} = \sum_{u\in U}|V_u| \le \beta\sum_{u\in U_d}|A\cap V_u| + W \le \beta k s + W$. With $C = \sum_u c_u \le s$, the bound $\log\binom{n+c-1}{c} \le c\log\frac{e(n+c)}{c}$ and the log-sum inequality give

$$\sum_{u\in U}\log\binom{|V_u|+c_u-1}{c_u} \le C\log\frac{e(V_{\text{tot}}+C)}{C} \le s\log\frac{e(V_{\text{tot}}+s)}{s} \le s\log\bigl(e(\beta k+1)\bigr) + \frac{(\log e)\,W}{\beta k+1},$$

because $C \mapsto C\log(e(V+C)/C)$ is increasing and $\log(a+x) \le \log a + x\log e/a$. Finally $\log(s(s+1)) \le s\log 3$, and $|V(U)| \le V_{\text{tot}}$ gives $\log\binom{|V(U)|}{s} \le s\log\frac{e|V(U)|}{s} \le s\log(e\beta k) + (\log e)W/(\beta k)$. Altogether

$$\log\frac{1}{Q_1(A)} \le \Lambda_\lambda + |A|\bigl(\log k + \beta + 1 + 2\log 3 + 2\log e + \log(\beta k+1) + \log(\beta k)\bigr) + \kappa_\beta W.$$

Summation. As in the proof of Theorem 2.3, $\sum_{A\neq\emptyset} P(A)W(A) = \sum_i \mathrm{Inf}_i(f)\, w_i\, \nu_i(E_i)$. By Lemma 2.2, $\nu_i(E_i) \le \min(1, k\max_s \tau_\beta(|V_{t_s}|))$, and $w_i\tau_\beta(m) \le T_\beta(w_i)$ for $m \ge w_i$, so $w_i\nu_i(E_i) \le \min(w_i, kT_\beta(w_i)) \le M_\beta(k)$. By Lemma 2.1, $\varepsilon\Lambda_\lambda \le 4p_0(\log e)\mu/k \le (\log e)\, I(f)$, and $h(\varepsilon) \le 2.886\, I(f)$ by Lemma 1.5. Inserting these bounds in (2), with Q_1 in place of the measure used there, and using $2.886 + \log e + 1 + 2\log 3 + 2\log e < 11.39$ proves the first claim.

For the second take $\beta = 8$, so that $D_8 = 0.3163\ldots$ and (4) gives $M_8(1) \le 3.17$ and $M_8(k) \le 1 + 2.2\log k$ for $k \ge 2$. Since $\log(8k+1) \le 3.17 + \log k$ and $\log(8k) = 3 + \log k$, the constant is at most $25.56 + 1.35\cdot 3.17 < 30$ for $k = 1$, and at most $25.56 + 3\log k + 1.18\,(1 + 2.2\log k) < 30 + 5.6\log k$ for $k \ge 2$. □

Optimising β for each k improves the constant of Theorem 2.5 to less than $20 + 4.44\log k$ for $k \le 10^6$. The bound of Theorem 2.3 is smaller for $k \le 2$. The reference measure does not need the activation of the dense terms, which is the source of the factor k in Theorem 2.3: once one term of a connected piece of the cover is activated, a term overlapping it is described by a variable of an already chosen term and an index among at most k terms, and the variables it brings are paid for only through the density of A in $V(U)$, which is at least $1/(\beta k + 1)$ up to the sparse terms.

3. Regular functions

Negating an input multiplies $\widehat{g}(A)$ by ± 1, so we may assume that f is monotone and that its coordinates are ordered by decreasing desirability: $f(x^{i\leftarrow 1, j\leftarrow 0}) \ge f(x^{i\leftarrow 0, j\leftarrow 1})$ whenever $i < j$. For $i \in [n]$ let $U_i = \{1, \ldots, i-1\}$.

Lemma 3.1. *For every i and every assignment z to the coordinates outside $U_i \cup \{i\}$, the set of $y \in \{0,1\}^{U_i}$ such that i is pivotal at (y, z) is an antichain of $\{0,1\}^{U_i}$.*

Proof. Suppose $y < y'$ both have i pivotal, choose $j \in U_i$ with $y_j = 0 < y'_j$, and let $y'' = y \oplus e_j \le y'$. By monotonicity and since $j < i$,

$$f(y', z, x_i = 0) \ge f(y'', z, x_i = 0) \ge f(y, z, x_i = 1) = 1,$$

so i is not pivotal at (y', z). □

Lemma 3.2 (Littlewood–Offord form of Sperner's theorem)**.** *If $\mathcal{A} \subseteq \{0,1\}^m$ is an antichain, then $|\mathcal{A}|\, 2^{-m} \le \binom{m}{\lfloor m/2\rfloor} 2^{-m} \le \frac{2}{\sqrt{\pi}}(m+1)^{-1/2}$ for every $m \ge 0$.*

Proof. The first inequality is Sperner's theorem [25]. For even $m = 2r \ge 2$, $\binom{2r}{r}4^{-r} \le (\pi r)^{-1/2} = (2/(\pi m))^{1/2}$. For odd $m = 2r+1 \ge 3$, $\binom{2r+1}{r}2^{-2r-1} = \binom{2r}{r}4^{-r}\frac{2r+1}{2r+2}$ and $(2r+1)^3 \le 2r(2r+2)^2$, which again gives $\le (2/(\pi m))^{1/2}$; for $m = 1$ the bound $1/2 \le (2/\pi)^{1/2}$ is direct. Finally $(2/(\pi m))^{1/2} \le \frac{2}{\sqrt{\pi}}(m+1)^{-1/2}$ for $m \ge 1$, and for $m = 0$ the claim reads $1 \le 2/\sqrt{\pi}$. □

Lemma 3.3 (Partial noise sensitivity of the derivative)**.** *Let $i \ge 2$, $N = i$, $0 < c \le N$ and $\rho = 1 - c/N$. Then*

$$\sum_T \widehat{D_i g}(T)^2\, \rho^{|T\cap U_i|} \le \frac{2}{\sqrt{\pi}}\, c^{-1/2}\, \mathrm{Inf}_i(f).$$

Proof. Write $d = D_i g$. Let x be uniform, let $R \subseteq U_i$ contain each $j \in U_i$ independently with probability c/N, and let y agree with x outside R and be uniform on R, independently of x. Then $\mathbb{E}[\chi_T(x)\chi_{T'}(y)] = \mathbf{1}[T = T']\rho^{|T\cap U_i|}$, so the left side equals $\mathbb{E}[d(x)d(y)] \le \mathbb{E}\big[|d(x)|\mathbb{P}[d(y) \ne 0 \mid x]\big]$. Given x and R, the points y_R with $d(y) \ne 0$ form an antichain of $\{0,1\}^R$ by Lemma 3.1 (the restriction of an antichain to a subcube is an antichain), so Lemma 3.2 gives $\mathbb{P}[d(y) \ne 0 \mid x, R] \le$

$\frac{2}{\sqrt{\pi}}(|R|+1)^{-1/2}$. The size $|R|$ has the law $\mathrm{Bin}(N-1, c/N)$, so $\mathbb{E}[1/(|R|+1)] = (1-(1-c/N)^N)/c \le 1/c$ and, by Jensen's inequality, $\mathbb{E}[(|R|+1)^{-1/2}] \le c^{-1/2}$. Finally $\mathbb{E}|d(x)| = \mathrm{Inf}_i(f)$. □

Lemma 3.4. *Let $B = 3\sqrt{3/\pi}$ and $L_1 = 2\ln B + 2 = 4.1511\ldots$. For every i with $\mathrm{Inf}_i(f) > 0$,*

$$\mathbb{E}_{T\sim\nu_i}\Big[\ln \frac{i}{|T\cap U_i|+1}\Big] \le L_1.$$

Proof. For $i = 1$ the left side is 0. Let $i \ge 2$, $N = i$ and $\kappa(T) = |T \cap U_i| \in [0, N-1]$. For real $K \ge 1$ apply Lemma 3.3 with $c = N/(3K)$, so that $\rho = 1 - 1/(3K)$:

$$\nu_i(\kappa \le K) \le \rho^{-K}\,\mathbb{E}_{\nu_i}\rho^{\kappa} \le \Big(1-\frac{1}{3K}\Big)^{-K}\frac{2}{\sqrt{\pi}}\Big(\frac{3K}{N}\Big)^{1/2} \le B\Big(\frac{K}{N}\Big)^{1/2},$$

because $-K\ln(1-\frac{1}{3K}) = \sum_{m\ge1} 3^{-m}K^{1-m}/m$ is nonincreasing in K and equals $\ln\frac{3}{2}$ at $K = 1$. Since $\ln(N/(\kappa+1)) \in [0, \ln N]$,

$$\mathbb{E}_{\nu_i}\ln\frac{N}{\kappa+1} = \int_0^{\ln N} \nu_i\big(\kappa+1 < Ne^{-u}\big)\,du \le \int_0^{\infty} \min\big(1, Be^{-u/2}\big)\,du = 2\ln B + 2,$$

using the previous bound with $K = Ne^{-u} \ge 1$. □

Lemma 3.5. $\mathbb{E}_S\big[\sum_{i\in S}\ln^+(i/|S|)\big] \le L_1\, I(f)$.

Proof. For $i \in S$ and $T = S\setminus\{i\}$ we have $|T\cap U_i|+1 \le \min(|S|, i)$, hence $\ln^+(i/|S|) \le \ln(i/(|T\cap U_i|+1))$. Since $\mathbb{P}[S = T\cup\{i\}] = \mathrm{Inf}_i(f)\nu_i(T)$, summing Lemma 3.4 with weights $\mathrm{Inf}_i(f)$ proves the claim. □

Lemma 3.6 (Rank-weighted reference measure)**.** *For every random subset S of $[n]$ and every $a > 1$,*

$$H(S \mid |S|) \le \frac{1}{\ln 2}\Big(\ln\frac{ea}{a-1}\,\mathbb{E}|S| + a\,\mathbb{E}\sum_{i\in S}\ln^+\frac{i}{|S|}\Big).$$

Proof. For $k \ge 1$ let $\theta_k(s) = \min(1, (k/s)^a)$ and let $\pi_k(A) = Z_k^{-1}\prod_{i\in A}\theta_k(i)$ on k-sets, where Z_k is the k-th elementary symmetric polynomial of $\theta_k(1),\ldots,\theta_k(n)$. Then $Z_k \le (\sum_s \theta_k(s))^k/k!$, $\sum_s\theta_k(s) \le k + \int_k^\infty (k/s)^a\,ds = \frac{ka}{a-1}$ and $k! \ge (k/e)^k$, so $\ln Z_k \le k\ln\frac{ea}{a-1}$. Gibbs' inequality on each level gives $H(S \mid |S| = k) \le \frac{1}{\ln 2}\mathbb{E}\big[\ln Z_k + a\sum_{i\in S}\ln^+(i/k) \;\big|\; |S| = k\big]$; average over k. □

Proof of Theorem 1.2. By the chain rule, Lemma 1.5, Lemma 3.6 and Lemma 3.5,

$$H(f) = H(|S|) + H(S \mid |S|) \le \Big(3.8854 + \frac{1 + \ln\frac{a}{a-1} + aL_1}{\ln 2}\Big) I(f).$$

The bracket is minimised at $a = \frac{1}{2}\big(1+\sqrt{1+4/L_1}\big) = 1.2006\ldots$, where the second summand equals $11.214\ldots$, giving $H(f) \le 15.1\, I(f)$. □

The same argument applies to Boolean functions of several regular functions, even when their desirability orders are unrelated.

Theorem 3.7. *Let $h_1,\ldots,h_m$ be regular Boolean functions on $\{0,1\}^n$, with arbitrary and possibly different desirability orders and orientations, and let $F : \{0,1\}^m \to \{0,1\}$ be arbitrary. Then $f = F(h_1,\ldots,h_m)$ satisfies*

$$H(f) \le (16.5 + 8.4\ln m)\, I(f).$$

In particular this holds for every intersection of m halfspaces, and for $x \mapsto G(w\cdot x)$ when $G : \mathbb{R} \to \{0,1\}$ changes sign at most m times.

Proof. For each l fix an orientation and a desirability order of h_l, let $r_l(i)$ be the position of i in the l-th order and let U_i^l be the set of the $r_l(i)-1$ coordinates that precede i in it. If i is pivotal for f at x, then it is pivotal for some h_l at x. Let $d_{i,l} = D_i g \cdot \mathbf{1}[\,l$ is the least index for which i is pivotal for $h_l\,]$; these functions have disjoint supports, $D_i g = \sum_l d_{i,l}$, and the support of $d_{i,l}$ lies in the pivotal set of i for h_l. Lemma 3.1 applied to h_l shows that this support, restricted to any subcube spanned by coordinates of U_i^l, is an antichain in that subcube. The proof of Lemma 3.3 uses nothing else, so it applies to $d_{i,l}$ with U_i^l in place of U_i, and the proof of Lemma 3.4 then gives, for the probability measure $\nu_{i,l}(T) = \widehat{d_{i,l}}(T)^2/\|d_{i,l}\|_2^2$ and $F_l(T) = \ln \frac{r_l(i)}{|T\cap U_i^l|+1}$,

$$\nu_{i,l}(F_l > u) \le \min(1, Be^{-u/2}) \quad (u \ge 0), \qquad \text{hence} \qquad \mathbb{E}_{\nu_{i,l}} e^{\lambda F_l} \le \frac{B^{2\lambda}}{1-2\lambda} \quad (0 < \lambda < \tfrac12).$$

As in the proof of Lemma 2.2, the Cauchy–Schwarz inequality gives $\nu_i \le \sum_l \nu_{i,l}$. For $G_i = \min_l F_l$, Jensen's inequality and this domination give, with $\lambda = \frac14$,

$$\mathbb{E}_{\nu_i} G_i \le \frac{1}{\lambda} \ln \mathbb{E}_{\nu_i} e^{\lambda G_i} \le \frac{1}{\lambda} \ln \sum_l \mathbb{E}_{\nu_{i,l}} e^{\lambda F_l} \le 4\ln m + 4\ln 2 + 2\ln B < 4\ln m + 4.93.$$

Since $\ln^+ \frac{r_l(i)}{|S|} \le F_l(S\setminus\{i\})$ for $i \in S$, summing with weights $\mathrm{Inf}_i(f)$ as in the proof of Lemma 3.5 gives $\mathbb{E}_S \sum_{i\in S} \min_l \ln^+ \frac{r_l(i)}{|S|} \le (4\ln m + 4.93)\, I(f)$.

Now repeat the proof of Lemma 3.6 with $\theta_k(i) = \max_l \min\bigl(1, (k/r_l(i))^a\bigr)$. Each order is a permutation of $[n]$, so $\sum_i \theta_k(i) \le mka/(a-1)$ and $\ln Z_k \le k\bigl(1 + \ln m + \ln \frac{a}{a-1}\bigr)$, while $\ln \frac{1}{\theta_k(i)} = a \min_l \ln^+ \frac{r_l(i)}{k}$. Together with Lemma 1.5,

$$H(f) \le \Bigl(3.886 + \frac{1 + \ln \frac{a}{a-1} + 4.93\, a + (1+4a)\ln m}{\ln 2}\Bigr) I(f),$$

and $a = 1.2$ gives the stated bound. For $G(w\cdot x)$, write f as a Boolean function of the m threshold functions $\mathrm{sgn}(w\cdot x - \theta_l)$ at the sign changes of G. □

When the orders of $h_1, \dots, h_m$ coincide, the supports of the pieces lie in a union of m antichains of the same predecessor cube, the bound of Lemma 3.2 is simply multiplied by m, and the constant improves to $3.8854 + \min_{a>1}\bigl(1 + \ln \frac{a}{a-1} + a(L_1 + 2\ln m)\bigr)/\ln 2 = 15.1 + O(\log m)$.

Remark 3.8. Theorem 3.7 applies to decision lists whose tests are m threshold functions, to unions and intersections of m halfspaces, and to DNF formulas with m terms, each term being a threshold function; for read-k DNF formulas with few terms it can be stronger than Theorem 1.1. It also applies to every function f that becomes regular after fixing the values of a set K of k coordinates (the order may depend on the values): f is a Boolean function of the k coordinates in K and of the 2^k restrictions $f(a,\cdot)$, $a \in \{0,1\}^K$, all of which are regular, so $H(f) \le \bigl(16.5 + 8.4\ln(k+2^k)\bigr) I(f)$.

Remark 3.9. The proof of Theorem 1.2 uses the order only through Lemma 3.1, which needs every coordinate of U_i to be at least as desirable as i, and through the bound $\sum_s \theta_k(s) \le ka/(a-1)$ in Lemma 3.6. Let f be monotone, let $\preceq$ be its desirability preorder, refined inside each class of equivalent coordinates by a fixed order, and let w be the largest number of pairwise incomparable coordinates. Take U_i to be the set of coordinates above i and $r(i) = |U_i| + 1$. Lemmas 3.1 and 3.3 to 3.5 hold with the same proofs. By Dilworth's theorem the coordinates form w chains, and the p-th element of a chain has $r \ge p$, so $\sum_i \min(1, (k/r(i))^a) \le wka/(a-1)$ and $\ln Z_k \le k\bigl(\ln \frac{ea}{a-1} + \ln w\bigr)$. Hence $H(f) \le (15.1 + \log w)\, I(f)$. A monotone function invariant under $S_{n_1} \times \cdots \times S_{n_w}$ acting on w blocks of coordinates has width at most w; for these functions,

with or without monotonicity, O'Donnell, Wright and Zhou [22] proved the better bound $H(f) \le (12.04 + \log w)\, I(f)$.

Remark 3.10. O'Donnell, Wright and Zhou [22, Theorem 5] proved $H(|S|) \le 3I(f)$ for every Boolean function. Using it in place of the second bound of Lemma 1.5, the constant of Theorem 1.2 becomes 14.22, and those of Theorem 3.7 and Remark 3.9 decrease accordingly. The function AND_n is a threshold function with $H/I \to 4$, so the optimal constant for regular functions is at least 4.

4. Unions of regular functions

The two mechanisms combine. In this section $f = h_1 \vee \cdots \vee h_m$, where h_l depends exactly on the variables in a set R_l, and every variable lies in at most k of the sets R_l.

Theorem 4.1. *If $h_1, \dots, h_m$ are regular and every variable is relevant to at most k of them, then $h_1 \vee \cdots \vee h_m$ and $h_1 \wedge \cdots \wedge h_m$ satisfy $H(f) \le (3.9 + 21.2k)\, I(f)$.*

The terms of a DNF formula are threshold functions, so Theorem 4.1 contains Theorems 1.1 and 1.2 up to the constants. It also applies to unions and intersections of halfspaces in which every variable has a nonzero weight in at most k of the halfspaces, with a constant that does not depend on their number.

Lemma 4.2. *Let $h_1, \dots, h_m$ be arbitrary Boolean functions, h_l depending only on the variables in R_l, with every variable in at most k of the sets R_l. For $f = h_1 \vee \cdots \vee h_m$ and $p_0 = \mathbb{P}[f = 0]$,*

$$p_0 \sum_l \mathbb{P}[h_l = 1] \le \frac{k}{4}\, I(f).$$

Proof. Fix l, let x be uniform and let x' be obtained from x by resampling the coordinates in R_l. Since $h_l(x')$ depends only on x'_{R_l}, which is independent of x, and $f \ge h_l$, $\mathbb{P}[f(x) = 0,\ f(x') = 1] \ge p_0 \mathbb{P}[h_l = 1]$. Now resample the coordinates of R_l one at a time. Every intermediate point is uniform, and resampling a coordinate j of a uniform point changes f from 0 to 1 with probability $\mathrm{Inf}_j(f)/4$. If $f(x) = 0$ and $f(x') = 1$, one of these steps does so, so the probability above is at most $\frac{1}{4} \sum_{j \in R_l} \mathrm{Inf}_j(f)$. Summing over l proves the lemma. □

In Lemma 2.2 the subcube structure of a term made the spectral sample fill the witnessing term uniformly. For a regular function we use instead that its pivotal set is small when the function is rarely true.

Lemma 4.3. *Let h be a Boolean function depending only on the variables in a set $R \ni j$, let $\alpha = \mathrm{Inf}_j(h)$, and let $d : \{0,1\}^{[n]\setminus\{j\}} \to \{-1, 0, 1\}$ vanish at every point at which j is not pivotal for h. For the probability measure $\nu(T) = \widehat{d}(T)^2 / \|d\|_2^2$ and every integer $0 \le \ell \le \ln(1/\alpha)$,*

$$\nu\big(|T \cap R| \le \ell\big) \le \varphi_\ell(\alpha), \qquad \text{where } \varphi_0(\alpha) = \alpha \text{ and } \varphi_\ell(\alpha) = \alpha\Big(\frac{2e \ln(1/\alpha)}{\ell}\Big)^\ell \text{ for } \ell \ge 1.$$

Proof. Write the argument of d as (y, z) with y the restriction to $R \setminus \{j\}$, put $d_z = d(\cdot, z)$ and $W^{\le \ell}[\phi] = \sum_{|U| \le \ell} \widehat{\phi}(U)^2$. Parseval's identity in z gives $\sum_{T:\, |T \cap R| \le \ell} \widehat{d}(T)^2 = \mathbb{E}_z W^{\le \ell}[d_z]$ and $\|d\|_2^2 = \mathbb{E}_z \|d_z\|_2^2$. Each d_z takes values in $\{-1, 0, 1\}$ and vanishes outside the set of y at which j is pivotal for h, a set of measure α, so $\|d_z\|_2^2 = \alpha_z \le \alpha$. For $\ell = 0$, $W^{\le 0}[d_z] = (\mathbb{E} d_z)^2 \le \alpha_z^2$. For $\ell \ge 1$ with $\alpha_z > 0$ put $\delta = \ell / (2 \ln(1/\alpha_z)) \le \frac{1}{2}$. By Hölder's inequality and the hypercontractive inequality $\|\phi\|_{1+1/\delta} \le \delta^{-\ell/2} \|\phi\|_2$ for ϕ of degree at most ℓ,

$$W^{\le \ell}[d_z] = \langle d_z, d_z^{\le \ell} \rangle \le \alpha_z^{1/(1+\delta)}\, \delta^{-\ell/2}\, W^{\le \ell}[d_z]^{1/2},$$

hence $W^{\le\ell}[d_z] \le \alpha_z^{2-2\delta}\delta^{-\ell} = \alpha_z^2\big(2e\ln(1/\alpha_z)/\ell\big)^{\ell}$. In both cases $W^{\le\ell}[d_z] \le \alpha_z\varphi_\ell(\alpha_z)$. The function φ_ℓ is nondecreasing on $(0,e^{-\ell}]$, which contains every α_z, so $\mathbb{E}_z W^{\le\ell}[d_z] \le \varphi_\ell(\alpha)\,\mathbb{E}_z\alpha_z$. □

Proof of Theorem 4.1. The complement of $h_1\wedge\cdots\wedge h_m$ is the union of the complements of the h_l, which are regular, and a function and its complement have the same spectral distribution; so it suffices to treat unions. For each l fix an orientation under which h_l is monotone and an order of R_l by decreasing desirability for h_l; let $r_l(j)$ be the position of $j\in R_l$ and U_j^l the set of the $r_l(j)-1$ variables before it. Put $\pi_l = \min(\mathbb{P}[h_l=1],\frac12)$, $\varepsilon = \operatorname{Var}(f)$, $a = 1.44$, $\beta = 5$, $c_1 = 1+\frac{a}{a-1}$ and $c_2 = 1+\frac{c_1}{\beta}$.

Reference measure. For $\kappa\ge 1$ let B_κ^l be the law of the random subset of R_l that contains each j independently with probability $\theta_\kappa^l(j) = \frac12\min\big(1,(\kappa/r_l(j))^a\big)$. Let L contain each index l independently with probability π_l, for each $l\in L$ choose $\kappa_l\ge 1$ with probability $2^{-\kappa_l}$ and then $A_l\sim B_{\kappa_l}^l$, all independently, and let $Q_1(A)$ be the probability that $\bigcup_{l\in L}A_l = A$. With $Q(\emptyset) = 1-\varepsilon$ and $Q(A) = \varepsilon Q_1(A)$ for $A\ne\emptyset$ we have $\sum_A Q(A)\le 1$, and $Q(A)>0$ whenever $P(A)>0$, so Gibbs' inequality gives

$$H(f) \le h(\varepsilon) + \sum_{A\ne\emptyset} P(A)\log\frac{1}{Q_1(A)}.$$

A choice for each A. Fix $A\ne\emptyset$ and put $\kappa_l = |A\cap R_l|$ and $\lambda_l = \frac{1}{\beta}\log\frac{1}{\pi_l}$; call l dense if $\kappa_l\ge\lambda_l$. For $j\in A$ and each l with $j\in R_l$ put $F_l^j = \ln\frac{r_l(j)}{|A\cap U_j^l|+1}\ge 0$ and

$$G_j(l) = \frac{a}{\ln 2}F_l^j + c_2\log\frac{1}{\pi_l}\,\mathbf{1}[\,l\text{ is not dense}\,],$$

let $\tau(j)$ minimise $G_j(l)$ over these l, and write $G_j = G_j(\tau(j))$. Let L_A consist of the dense indices and the $\tau(j)$, $j\in A$, and for $l\in L_A$ let $A_l = \{j\in A:\tau(j)=l\}$. Then $A = \bigcup_{l\in L_A}A_l$ and $\kappa_l\ge 1$ for $l\in L_A$. Using $\log\frac{1}{1-x}\le 2x$ for $0\le x\le\frac12$ and $\sum_{j\in R_l}\theta_\kappa^l(j)\le\frac12\big(\kappa+\int_\kappa^\infty(\kappa/r)^a\,dr\big) = \frac{\kappa a}{2(a-1)}$,

$$\log\frac{1}{B_\kappa^l(A_l)} \le \sum_{j\in A_l}\Big(1+a\log^+\frac{r_l(j)}{\kappa}\Big) + \frac{\kappa a}{a-1},$$

and for $j\in A_l$ we have $\kappa_l\ge|A\cap U_j^l|+1$, so that $\log^+(r_l(j)/\kappa_l)\le F_l^j/\ln 2$. With $\Lambda = \sum_l\log\frac{1}{1-\pi_l}$ this gives

$$\log\frac{1}{Q_1(A)} \le \Lambda + \sum_{l\in L_A}\Big(\log\frac{1}{\pi_l}+c_1\kappa_l\Big) + \sum_{j\in A}\Big(1+\frac{a}{\ln 2}F_{\tau(j)}^j\Big).$$

For dense l, $\log\frac{1}{\pi_l}\le\beta\kappa_l$. A nondense $l\in L_A$ equals $\tau(j)$ for some $j\in A$ and satisfies $\log\frac{1}{\pi_l}+c_1\kappa_l < c_2\log\frac{1}{\pi_l}$. As $\sum_l\kappa_l\le k|A|$,

$$\log\frac{1}{Q_1(A)} \le \Lambda + (\beta+c_1)\,k|A| + \sum_{j\in A}(1+G_j).$$

The pieces. For j with $\operatorname{Inf}_j(f)>0$ let $l_1,\dots,l_r$ ($r\le k$) be the indices with $j\in R_{l_s}$. If j is pivotal for f at x, then it is pivotal for some h_{l_s} at x. As in the proof of Theorem 3.7, split D_jg according to the least such s into pieces d_s with disjoint supports, d_s vanishing where j is not pivotal for h_{l_s}, and let $\nu_{j,s}$ be the normalised spectral measure of d_s. The Cauchy–Schwarz

inequality gives $\nu_j \le \sum_s \nu_{j,s}$, and $G_j \le G_j(l_s)$ for every s, so with $A = T \cup \{j\}$,

$$\mathbb{E}_{T\sim\nu_j} G_j \le \sum_{s=1}^{r} \Big(\frac{a}{\ln 2} \mathbb{E}_{\nu_{j,s}} F^j_{l_s} + c_2 \log \frac{1}{\pi_{l_s}} \, \nu_{j,s}\big(l_s \text{ is not dense}\big) \Big).$$

The proof of Theorem 3.7 gives $\mathbb{E}_{\nu_{j,s}} F^j_{l_s} \le L_1$. Put $\lambda = \lambda_{l_s}$. If $\lambda \le 1$, then l_s is dense because $j \in A \cap R_{l_s}$. Otherwise $\pi_{l_s} = 2^{-\beta\lambda} < \frac{1}{2}$, and l_s is not dense exactly when $|T \cap R_{l_s}| \le \ell := \lceil \lambda \rceil - 2$. At a point where j is pivotal for h_{l_s}, one of the two neighbours has $h_{l_s} = 1$, so $\mathrm{Inf}_j(h_{l_s}) \le 2\mathbb{P}[h_{l_s} = 1] = 2^{1-\beta\lambda}$; as $\ell \le (\beta\lambda - 1)\ln 2$, Lemma 4.3 gives

$$\log \frac{1}{\pi_{l_s}} \, \nu_{j,s}\big(l_s \text{ is not dense}\big) \le \beta\lambda \, \varphi_\ell\big(2^{1-\beta\lambda}\big) \le K := \max_{\ell \ge 0} 5(\ell+1)\, \varphi_\ell\big(2^{-5\ell-4}\big) = 0.6624\ldots;$$

here we used that for fixed ℓ the middle expression decreases in $\lambda \in (\ell+1, \ell+2]$, since its logarithmic derivative is $\frac{1}{\lambda} - \beta \ln 2 \, (1 - \ell/((\beta\lambda - 1)\ln 2)) < 0$, and the maximum is attained at $\ell = 1$.

Conclusion. Since $P(A) = \mathrm{Inf}_j(f)\, \nu_j(A \setminus \{j\})$ for $j \in A$, summing over A gives

$$H(f) \le h(\varepsilon) + \varepsilon\Lambda + (\beta + c_1) k \, I(f) + I(f) + k\Big(\frac{aL_1}{\ln 2} + c_2 K \Big) I(f).$$

By Lemma 1.5, $h(\varepsilon) \le 2.8854\, I(f)$, and by Lemma 4.2, $\varepsilon\Lambda \le 4p_0 \cdot 2\sum_l \pi_l \le 2k\, I(f)$. The coefficient of k is $2 + 5 + 4.273 + 8.624 + 1.855 \cdot 0.6624 < 21.2$. □

The device of Section 2.1 also applies here, with the witnessing terms replaced by the functions h_l and the uniform subsets of $V(U)$ replaced by rank laws.

Theorem 4.4. *Under the hypotheses of Theorem 4.1,* $H(f) \le (54 + 12.8 \log k)\, I(f)$.

Proof. Keep the notation of the proof of Theorem 4.1, but put $a = 1.44$, $\beta = 10$ and $\lambda = 1/k$, and let $\theta^l_s(i) = \min(1, (s/r_l(i))^a)$ for $i \in R_l$ and Z^l_s the s-th elementary symmetric function of the numbers $\theta^l_s(i)$, $i \in R_l$; as in the proof of Lemma 3.6, $\log Z^l_s \le s \log \frac{ea}{a-1}$.

Reference measure. (a) Choose a set R of indices, containing each l independently with probability $\lambda\pi_l/(1 + \lambda\pi_l)$. (b) Put the indices of R into a queue. When l is taken from the queue, choose $s \ge 1$ with probability $1/(s(s+1))$ and an s-element subset $B_l \subseteq R_l$ with probability $\prod_{i\in B_l} \theta^l_s(i)/Z^l_s$; then choose $c \ge 0$ with probability $1/((c+1)(c+2))$, a multiset of c elements of B_l uniformly, and for each chosen element j an index l' with $j \in R_{l'}$, uniformly among the at most k such indices; append these indices to the queue. (c) When the queue is empty, output $\bigcup_l B_l$. Let $Q_1(A)$ be the probability that the output is A.

A run producing A. Let $A \ne \emptyset$ with $P(A) > 0$. Define F^j_l, the dense indices and $G_j(l)$ as in the proof of Theorem 4.1, with c_2 replaced by 1, let $\tau(j)$ minimise $G_j(l)$ over the l with $j \in R_l$, put $A_l = \{j \in A : \tau(j) = l\}$ and let $\mathcal{F}$ be the set of l with $A_l \ne \emptyset$. Consider the directed graph on $\mathcal{F}$ with an edge $l \to l'$ when $l \ne l'$ and $A_l \cap R_{l'} \ne \emptyset$, choose one index in each strongly connected component that receives no edge from another component, and let R be the set of these roots. Every index of $\mathcal{F}$ is reached from a root; take a breadth-first spanning forest along the edges, attach each non-root l' to its parent l through an element of $A_l \cap R_{l'}$, and let $B_l = A_l$ for $l \in \mathcal{F}$. This is a run that outputs A, with $|R| + \sum_l c_l = |\mathcal{F}| \le |A|$, where c_l is the number of children of l.

The sets $A \cap R_r$, $r \in R$, are pairwise disjoint. Indeed, if $j \in A \cap R_r \cap R_{r'}$ with $r \ne r'$ and $l = \tau(j)$, then either $l \notin \{r, r'\}$ and both $l \to r$ and $l \to r'$ are edges, or $l = r$ and $r \to r'$ is an edge, or $l = r'$ and $r' \to r$ is an edge; in each case two roots lie in one strong component or a root's component receives an edge from another one, which the choice of roots excludes.

Cost of the run. Let $\kappa_l = |A \cap R_l|$ and $N_j = \log \frac{1}{\pi_{\tau(j)}} \mathbf{1}[\tau(j) \text{ is not dense}]$, so that $G_j := G_j(\tau(j)) = \frac{a}{\ln 2} F^j_{\tau(j)} + N_j$. A dense root has $\log \frac{1}{\pi_r} \le \beta \kappa_r$, and a root that is not dense is $\tau(j)$ for some j; by the disjointness just proved, $\sum_{r \in R} \log \frac{1}{\lambda \pi_r} \le |R| \log k + \beta |A| + \sum_{j \in A} N_j$. For $i \in A_l$ with $s = |A_l|$, write $u = |A \cap U^l_i|$; then $u + 1 \le \kappa_l$ and $\ln^+(r_l(i)/s) \le F^i_l + \ln(\kappa_l / s)$, so by the log-sum inequality and $\sum_l \kappa_l \le k|A|$,

$$\sum_{l \in \mathcal{F}} \sum_{i \in A_l} \log \frac{1}{\theta^l_{|A_l|}(i)} \le \frac{a}{\ln 2} \sum_{j \in A} F^j_{\tau(j)} + a \sum_l |A_l| \log \frac{\kappa_l}{|A_l|} \le \frac{a}{\ln 2} \sum_{j \in A} F^j_{\tau(j)} + a|A| \log k.$$

The choices of s, of c, of the multisets and of the indices of the children cost at most $|A| \log 3$, $|A|(1 + \log 3)$, $|A| \log(2e)$ and $(|\mathcal{F}| - |R|) \log k$ in total, as in the proof of Theorem 2.5, and together with $|R| \log k$ from the roots the indices cost $|\mathcal{F}| \log k \le |A| \log k$. Hence

$$\log \frac{1}{Q_1(A)} \le \Lambda_\lambda + |A| \Big((1 + a) \log k + \beta + 1 + 2 \log 3 + \log \frac{ea}{a - 1} + \log(2e) \Big) + \sum_{j \in A} G_j,$$

with $\Lambda_\lambda = \sum_l \log(1 + \lambda \pi_l) \le (\log e) \sum_l \pi_l / k$.

The pieces. Let d_s, $\nu_{j,s}$ and l_s be as in the proof of Theorem 4.1, so that $\nu_j \le \sum_s \nu_{j,s}$ and $G_j \le G_j(l_s)$ for every s. Put $b_s = \log \frac{1}{\pi_{l_s}}$. The proof of Theorem 3.7 gives $\nu_{j,s}(F^j_{l_s} > v) \le B e^{-v/2}$, and the argument of the proof of Theorem 4.1 gives $\nu_{j,s}(l_s \text{ is not dense}) \le g(b_s)$, where $g(b) = \sup\{\varphi_{\ell(b')}(2^{1-b'}) : b' \ge b,\ b' > \beta\}$ and $\ell(b') = \lceil b'/\beta \rceil - 2$. On each interval $(10(\ell + 1), 10(\ell + 2)]$ the logarithm of $2^{0.44b} \varphi_\ell(2^{1-b})$ decreases, and its limits at the left end points are $-4.6, -4.04, -4.26, -4.56, \ldots$ for $\ell = 0, 1, 2, 3, \ldots$, decreasing from $\ell = 1$ on, so $g(b) \le C_g 2^{-\gamma b}$ with $\gamma = 0.44$ and $C_g = 2^{-4.04}$. Since $\{G_j(l_s) > u\} \subseteq \{\frac{a}{\ln 2} F^j_{l_s} > \frac{u}{2}\} \cup \{l_s \text{ not dense}, b_s > \frac{u}{2}\}$,

$$\nu_j(G_j > u) \le \min\big(1,\ kBe^{-u \ln 2/(4a)} + k\, C_g 2^{-\gamma u/2}\big),$$

and integrating, with $\min(1, x + y) \le \min(1, 2x) + \min(1, 2y)$ and $2C_g < 1$,

$$\mathbb{E}_{\nu_j} G_j \le \frac{4a}{\ln 2} \big(\ln(2kB) + 1 \big) + \frac{2}{\gamma} \big(\log k + \log e \big).$$

Conclusion. Sum over A as before. We have $h(\varepsilon) \le 2.886\, I(f)$, and $\varepsilon \Lambda_\lambda \le (\log e)\, I(f)$ because $\varepsilon \le 4p_0$ and Lemma 4.2 gives $4p_0 \sum_l \pi_l \le k\, I(f)$. The constant is at most $53.7 + (1 + a + 4a + 2/\gamma) \log k < 54 + 12.8 \log k$. □

The bound of Theorem 4.1 is smaller for $k \le 3$. Optimising a and β for each k and integrating the tail bounds exactly gives a constant below 61 for $k \le 10$ and below 108 for $k \le 1000$.

The two mechanisms also combine with Theorem 3.7: the functions h_l may themselves be Boolean functions of several regular functions.

Theorem 4.5. *Let $f = h_1 \vee \cdots \vee h_m$ or $f = h_1 \wedge \cdots \wedge h_m$, where every h_l is a Boolean function of at most M regular functions, whose desirability orders are arbitrary, and every variable is relevant to at most k of the functions h_l. Then*

$$H(f) \le (54 + 12.8 \log k + 6.8 \log M)\, I(f) \qquad \text{and} \qquad H^+(f) \le (93 + 23.1 \log k + 12.6 \log M)\, J(f).$$

For instance, unions and intersections of polytopes with at most M facets in which every variable has a nonzero weight in at most k of the polytopes satisfy the conjecture with a constant that does not depend on their number, and so do read-k DNF formulas in which every term is replaced by a Boolean function of at most M threshold functions on its variables. A Boolean function of w variables is a Boolean function of the w dictators x_j, which are regular, so a union $h_1 \vee \cdots \vee h_m$ in which every h_l depends on at most w variables and every variable is relevant to

at most k of the h_l satisfies $H(f) \le (54 + 12.8\log k + 6.8\log w)\, I(f)$, whatever the functions h_l are.

Proof. A complement of a Boolean function of M regular functions is again one, so it suffices to treat unions. Write $h_l = F_l(h_{l,1}, \dots, h_{l,M})$ with $h_{l,r}$ regular, depending on the variables of a set $R_{l,r} \subseteq R_l$, and let $r_{l,r}(i)$ be the position of i in a desirability order of $h_{l,r}$ and $U_i^{l,r}$ the set of variables before it. The proof of Theorem 4.4 applies with three changes.

(i) In the reference measure put $\theta_s^l(i) = \max_r \min\bigl(1, (s/r_{l,r}(i))^a\bigr)$, the maximum over the r with $i \in R_{l,r}$. Then $\sum_i \theta_s^l(i) \le Msa/(a-1)$, and as in the proof of Lemma 3.6, $\log Z_s^l \le s \log \frac{eMa}{a-1}$.

(ii) In the run put $F_l^j = \min_r \ln\bigl(r_{l,r}(j)/(|A \cap U_j^{l,r}| + 1)\bigr)$, the minimum over the r with $j \in R_{l,r}$. For $i \in A_l$ and the r attaining this minimum, $|A \cap U_i^{l,r}| + 1 \le \kappa_l$, so $\ln^+(r_{l,r}(i)/|A_l|) \le F_l^i + \ln(\kappa_l/|A_l|)$ and the rank costs are bounded as before. The per-set bound gains the term $|A| \log M$ from (i).

(iii) If j is pivotal for h_{l_s}, it is pivotal for some $h_{l_s,r}$. Splitting the piece d_s further according to the least such r gives at most M pieces, each vanishing where j is not pivotal for a regular function, so the Cauchy–Schwarz inequality and the proof of Theorem 3.7 give $\nu_{j,s}(F_{l_s}^j > v) \le MBe^{-v/2}$, while the bound on $\nu_{j,s}(l_s \text{ is not dense})$, which uses only Lemma 4.3, is unchanged. Hence $\mathbb{E}_{\nu_j} G_j \le \frac{4a}{\ln 2}\bigl(\ln(2kMB) + 1\bigr) + \frac{2}{\gamma}(\log k + \log e)$.

The path lemma (Lemma 4.2) holds for arbitrary h_l. The constant becomes $53.7 + 12.75\log k + (1+4a)\log M$ with $a = 1.44$. For H^+ the proof of Theorem 5.4(d) applies with the same changes. In its second case $\mathrm{Inf}_j(h_{l_s}) > 1/(2k)$, so some $h_{l_s,r}$ has $\mathrm{Inf}_j(h_{l_s,r}) > 1/(2kM)$; since the influences of a regular function do not increase along its order and their squares sum to at most 1, this gives $r_{l_s,r}(j) < 4k^2M^2$ and $G_j < a\log(4k^2M^2) + \log(4k)$ pointwise. The constant becomes $92.7 + 23.05\log k + (1+8a)\log M$. □

5. The strengthened inequality and composition

Recall $H^+(f) = \sum_{A \ne \emptyset} P(A)\log(1/P(A))$ from (1), and put

$$W(f) = \sum_{|A| \ge 2} P(A), \qquad J(f) = \sum_{|A| \ge 2} |A|\, P(A) = \sum_{i=1}^{n} \mathrm{Var}(D_i g).$$

The second expression for $J(f)$ holds because $\sum_{A \ni i,\, A \ne \{i\}} P(A) = \mathrm{Inf}_i(f) - \widehat{D_i g}(\emptyset)^2 = \mathrm{Var}(D_i g)$; note that $\widehat{D_i g}(\emptyset) = \widehat{g}(\{i\})$. Since $|A| \le 2(|A|-1)$ when $|A| \ge 2$, we have $J(f) \le 2\bigl(I(f) - \mathrm{Var}(f)\bigr)$, so it suffices to bound $H^+(f)$ by a multiple of $J(f)$. The next two lemmas hold for every Boolean function.

Lemma 5.1. *For every $i \in [n]$,* $P(\{i\}) \log \dfrac{1}{P(\{i\})} \le \dfrac{2}{\ln 2}\, \mathrm{Var}(D_i g)$.

Proof. Let $b = |\widehat{g}(\{i\})| = |\mathbb{E} D_i g|$. Then $b \le \mathbb{E}|D_i g| = \mathrm{Inf}_i(f)$, so $\mathrm{Var}(D_i g) = \mathrm{Inf}_i(f) - b^2 \ge b - b^2$. Since $\ln(1/b) \le 1/b - 1$, we get $P(\{i\})\log(1/P(\{i\})) = \frac{2}{\ln 2}\, b^2 \ln \frac{1}{b} \le \frac{2}{\ln 2}(b - b^2)$. □

The constant in Lemma 5.1 cannot be improved: for $f = x_1 \wedge \neg(x_2 \wedge \cdots \wedge x_w)$ and $i = 1$ the ratio of the two sides tends to $2/\ln 2$ as $w \to \infty$.

Lemma 5.2. $W(f) \log \dfrac{1}{W(f)} \le 8\, J(f)$.

Proof. Write $W = W(f)$, $J = J(f)$, $W_1 = \sum_i P(\{i\})$ and $\alpha = \min(p, 1-p)$. We use four facts. (i) $J \ge \sum_i \mathrm{Inf}_i(f)\bigl(1 - \mathrm{Inf}_i(f)\bigr)$, since $\mathrm{Var}(D_i g) = \mathrm{Inf}_i(f) - \widehat{g}(\{i\})^2$ and $|\widehat{g}(\{i\})| \le \mathrm{Inf}_i(f)$. (ii) $W \le J/2$ and $W \le \mathrm{Var}(f) \le 4\alpha$. (iii) Every edge in direction i along which f changes has

an endpoint in the smaller of $f^{-1}(0)$ and $f^{-1}(1)$, so $\mathrm{Inf}_i(f) \le 2\alpha$. Hence $W_1 \le \sum_i \mathrm{Inf}_i(f)^2 \le 2\alpha\, I(f)$ and $J = I(f) - W_1 \ge (1-2\alpha)I(f)$. (iv) $I(f) \ge 2\alpha \log(1/\alpha)$. Recall also that $x\log(1/x)$ is increasing on $(0, 1/e]$ and never exceeds $(\log e)/e < 0.531$.

Case $\alpha \le 1/4$. By (iii) and (iv), $J \ge I(f)/2 \ge \alpha\log(1/\alpha)$. If $4\alpha \le 1/e$, then (ii) gives $W\log(1/W) \le 4\alpha \log\frac{1}{4\alpha} \le 4J$. If $4\alpha > 1/e$, then $J \ge \alpha\log(1/\alpha) \ge 0.316$, so $W\log(1/W) < 0.531 < 1.7\,J$.

Case $\alpha > 1/4$ *and* $J \ge \delta := 0.024$. Put $w_0 = 0.012$. If $W \ge w_0$, then (ii) gives $W\log(1/W) \le \frac{J}{2}\log\frac{1}{w_0} < 3.2\,J$. If $W < w_0$, then $W\log(1/W) \le w_0\log(1/w_0) < 3.2\,\delta \le 3.2\,J$.

Case $\alpha > 1/4$ *and* $J < \delta$. Now $\mathrm{Var}(f) > 3/4$, so $I(f) \ge \mathrm{Var}(f) > 3/4$, while $I(f) = W_1 + J < 1+\delta$. By (i), every coordinate has $\mathrm{Inf}_i(f)(1-\mathrm{Inf}_i(f)) < \delta$, hence $\mathrm{Inf}_i(f) < 2\delta$ or $\mathrm{Inf}_i(f) > 1-2\delta$. Coordinates of the first kind satisfy $\mathrm{Inf}_i(f) \le \mathrm{Inf}_i(f)(1-\mathrm{Inf}_i(f))/(1-2\delta)$, so their total influence is less than $\delta/(1-2\delta) < 3/4$. Therefore a coordinate i^* of the second kind exists, and it is unique because $2(1-2\delta) > 1+\delta$. After negating x_{i^*} if necessary, $b := \widehat{g}(\{i^*\}) = \mathbb{E}D_{i^*}g$ satisfies $b^2 = \mathrm{Inf}_{i^*}(f) - \mathrm{Var}(D_{i^*}g) > 1-3\delta$, so $b > 1-3\delta$.

Write $g = A + \chi_{\{i^*\}}B$ with $A = \mathbb{E}_{i^*}g$ and $B = D_{i^*}g$. These are functions of the other variables with values in $\{-1,0,1\}$, and exactly one of them vanishes at each point. We have $\widehat{g}(T) = \widehat{A}(T)$ for $T \not\ni i^*$ and $\widehat{g}(T\cup\{i^*\}) = \widehat{B}(T)$. Let $D = 1-B \in \{0,1,2\}$ and let $E = \{D \ne 0\}$, of density η. From $b = 1-\mathbb{E}D$ and $\mathbb{E}D \ge \eta$ we get $\eta \le 1-b < 3\delta$. The sets with at least two elements are the sets $T \not\ni i^*$ with $|T| \ge 2$ and the sets $T\cup\{i^*\}$ with $T \ne \emptyset$; since $\|A\|_2^2 = \mathbb{P}[B=0] = \mathbb{P}[D=1]$,

$$W \le \|A\|_2^2 + \mathrm{Var}(B) \le \mathbb{P}[D=1] + \mathbb{E}[D^2] \le 4\eta.$$

On the other hand $J \ge \sum_{T\ne\emptyset}(|T|+1)\widehat{B}(T)^2 \ge \sum_T |T|\widehat{D}(T)^2 = \sum_j \mathbb{E}[(D_jD)^2]$. If an edge in direction j leaves E, then $|D_jD| \ge \frac12$ at both of its endpoints. Hence $\sum_j \mathbb{E}[(D_jD)^2]$ is at least $\frac14$ of the total influence of the indicator of E, which is at least $2\eta\log(1/\eta)$ by edge isoperimetry. Thus $J \ge \frac{\eta}{2}\log\frac{1}{\eta}$, and since $4\eta < 12\delta < 1/e$,

$$W\log\frac{1}{W} \le 4\eta\log\frac{1}{4\eta} \le 4\eta\log\frac{1}{\eta} \le 8J. \qquad \square$$

Proposition 5.3. *Let f be a Boolean function and $Q \ge 0$ a function on the subsets of $[n]$ with $\sum_{|A|\ge2} Q(A) \le 1$ and $Q(A) > 0$ whenever $|A| \ge 2$ and $P(A) > 0$. Then*

$$H^+(f) \le \Big(\frac{2}{\ln 2} + 8\Big)J(f) + \sum_{|A|\ge2} P(A)\log\frac{1}{Q(A)}.$$

Proof. Split $H^+(f)$ into the contribution of the singletons and that of the sets with at least two elements. By Lemma 5.1, the first is at most $\frac{2}{\ln 2}\sum_i \mathrm{Var}(D_ig) = \frac{2}{\ln 2}J(f)$. If $W = W(f) > 0$, Gibbs' inequality applied to the probability distribution $P(A)/W$ on $\{A : |A| \ge 2\}$ gives $\sum_{|A|\ge2} P(A)\log\frac{1}{P(A)} \le W\log\frac1W + \sum_{|A|\ge2} P(A)\log\frac{1}{Q(A)}$, and Lemma 5.2 bounds $W\log(1/W)$. $\square$

Theorem 5.4. (a) *If f is computed by a read-k DNF or CNF formula, then $H^+(f) \le \min(10.9+14k,\ 44+8.2\log k)\,J(f)$.*
(b) *If f is regular, then $H^+(f) \le 30\,J(f)$.*
(c) *If $f = F(h_1,\dots,h_m)$ as in Theorem 3.7, then $H^+(f) \le (33+15.3\ln m)\,J(f)$.*
(d) *If f is a union or an intersection of regular functions as in Theorem 4.1, then $H^+(f) \le \min(11.9+33k,\ 93+23.1\log k)\,J(f)$.*

Since $J(f) \le 2(I(f)-\mathrm{Var}(f))$, Theorem 5.4 implies Theorem 1.3.

Proof of Theorem 5.4(a). By duality, as in the proof of Theorem 1.1, it suffices to treat DNF formulas. Take $Q = Q_1$ from the proof of Theorem 2.3, a probability measure, and $\beta = 3$. For $|A| \geq 2$ and $i \in A$ the set $A \setminus \{i\}$ is nonempty, so the cover used in that proof gives

$$\sum_{|A|\geq 2} P(A) \log \frac{1}{Q_1(A)} \leq W(f)\,\Lambda + 2\beta k\, J(f) + 2 \sum_i \mathrm{Inf}_i(f)\, w_i\, \nu_i(E_i \setminus \{\emptyset\}).$$

The vacuum term. Recall $\Lambda \leq 2\mu$. If $p_0 \geq 1/4$, then $\mu \leq k\, I(f)$ by Lemma 2.1, and $W(f)I(f) = W(f)W_1 + W(f)J(f) \leq \frac{3}{2} J(f)$ by fact (ii) in the proof of Lemma 5.2, so $W(f)\Lambda \leq 3k\, J(f)$. If $p_0 < 1/4$, then $W(f) \leq \mathrm{Var}(f) \leq 4p_0$ and $J(f) \geq I(f)/2$ by fact (iii), so $W(f)\Lambda \leq 8p_0\mu \leq 2k\, I(f) \leq 4k\, J(f)$.

Sparse elements. We show $w_i\, \nu_i(E_i \setminus \{\emptyset\}) \leq \max(\log k + 2,\, 2kC_\beta)\, \nu_i(T \neq \emptyset)$, where $\nu_i(T \neq \emptyset) = \mathrm{Var}(D_i g)/\mathrm{Inf}_i(f)$. If $w_i < \log k + 2$ this is clear. Otherwise every term through i has at least $\log k + 2$ variables. In the notation of the proof of Lemma 2.2, $\widehat{d_s}(\emptyset) = 2^{-|W_s|}\widehat{\psi_s}(\emptyset)$ and $\|d_s\|_2^2 = 2^{-|W_s|}\|\psi_s\|_2^2$, so $|\widehat{d_s}(\emptyset)| \leq 2^{-|W_s|/2}\|d_s\|_2$, and the Cauchy–Schwarz inequality gives

$$\nu_i(\emptyset) = \frac{\widehat{D_i g}(\emptyset)^2}{\mathrm{Inf}_i(f)} \leq \sum_s 2^{-|W_s|} \leq k\, 2^{-\log k - 1} = \frac{1}{2}.$$

Hence $\nu_i(T \neq \emptyset) \geq \frac{1}{2}$, while $w_i \nu_i(E_i) \leq kC_\beta$ as in the proof of Theorem 2.3. Summing with weights $\mathrm{Inf}_i(f)$, the sparse term is at most $2\max(\log k + 2,\, 2kC_\beta)\, J(f)$.

With $\beta = 3$ and $C_3 < 0.949$, the three terms add up to at most $\big(4k + 6k + 2\max(\log k + 2, 1.898k)\big) J(f) \leq 14k\, J(f)$, and Proposition 5.3 gives $H^+(f) \leq (2/\ln 2 + 8 + 14k)\, J(f)$.

For the logarithmic bound take $Q = Q_1$ from the proof of Theorem 2.5. The bound on $\log(1/Q_1(A))$ obtained there holds for every $A \neq \emptyset$ and has no term independent of $|A|$ other than Λ_λ, so

$$\sum_{|A|\geq 2} P(A) \log \frac{1}{Q_1(A)} \leq W(f)\Lambda_\lambda + K_\beta\, J(f) + \kappa_\beta \sum_i \mathrm{Inf}_i(f)\, w_i\, \nu_i(E_i \setminus \{\emptyset\}),$$

with $K_\beta = \log k + \beta + 1 + 2\log 3 + 2\log e + \log(\beta k + 1) + \log(\beta k)$. As in the vacuum step above, with $\Lambda_\lambda \leq (\log e)\mu/k$ in place of $\Lambda \leq 2\mu$: if $p_0 \geq \frac{1}{4}$ then $W(f)\Lambda_\lambda \leq (\log e) W(f) I(f) \leq \frac{3}{2}(\log e) J(f)$, and if $p_0 < \frac{1}{4}$ then $W(f)\Lambda_\lambda \leq 4p_0(\log e)\mu/k \leq (\log e) I(f) \leq 2(\log e) J(f)$. The sparse term is at most $\max(\log k + 2,\, 2M_\beta(k))\, J(f)$ by the argument above, with $w_i \nu_i(E_i) \leq M_\beta(k)$ from the proof of Theorem 2.5 in place of kC_β. Hence

$$H^+(f) \leq \Big(\frac{2}{\ln 2} + 8 + 2\log e + K_\beta + \kappa_\beta \max\big(\log k + 2,\, 2M_\beta(k)\big)\Big) J(f),$$

and $\beta = 8$ with (4) gives at most $35.00 + 8.48 < 44$ for $k = 1$ and at most $37.35 + 8.15 \log k$ for $k \geq 2$. □

Proof of Theorem 5.4(b). As in Section 3, assume that f is monotone with coordinates ordered by decreasing desirability. Then $D_i g \geq 0$, so $\widehat{g}(\{i\}) = \mathrm{Inf}_i(f)$ and $\nu_i(\emptyset) = \mathrm{Inf}_i(f)$. Influences decrease along the order: if $i < j$ and z denotes the remaining coordinates, then $\mathrm{Inf}_i(f) - \mathrm{Inf}_j(f) = \mathbb{E}_z\big[f(x_i{=}1, x_j{=}0, z) - f(x_i{=}0, x_j{=}1, z)\big] \geq 0$. Since $\sum_i \mathrm{Inf}_i(f)^2 = \sum_i P(\{i\}) \leq 1$, a coordinate with $\mathrm{Inf}_i(f) > 1/2$ has $i \leq 3$.

We claim that

$$\sum_{|A|\geq 2} P(A) \sum_{i\in A} \ln^+ \frac{i}{|A|} \leq 2L_1\, J(f).$$

The left side equals $\sum_i \mathrm{Inf}_i(f)\, \mathbb{E}_{\nu_i}\big[\ln^+ \frac{i}{|T|+1}\, \mathbf{1}[T \neq \emptyset]\big]$, and $\ln^+ \frac{i}{|T|+1} \leq \ln \frac{i}{|T\cap U_i|+1}$. If $\mathrm{Inf}_i(f) \leq \frac{1}{2}$, then $\nu_i(T \neq \emptyset) \geq \frac{1}{2}$ and Lemma 3.4 bounds the expectation by $L_1 \leq 2L_1\, \nu_i(T \neq \emptyset)$. If

$\mathrm{Inf}_i(f) > \frac{1}{2}$, then $i \le 3$ and the expectation is at most $(\ln 3)\,\nu_i(T \neq \emptyset)$. Summing with weights $\mathrm{Inf}_i(f)$ proves the claim.

Now let $Q(A) = 2^{-(|A|-1)}\pi_{|A|}(A)$ for $|A| \ge 2$, with π_k as in the proof of Lemma 3.6. Then $\sum_{|A|\ge 2} Q(A) = 1$ and, by the bound on Z_k in that proof, $\log \frac{1}{Q(A)} \le |A|\big(1 + \log \frac{ea}{a-1}\big) + \frac{a}{\ln 2}\sum_{i\in A} \ln^+ \frac{i}{|A|}$. By Proposition 5.3 and the claim,

$$H^+(f) \le \Big(\frac{2}{\ln 2} + 9 + \log\frac{ea}{a-1} + \frac{2aL_1}{\ln 2}\Big) J(f),$$

and at $a = 1.1087$ the bracket is less than 30. □

Proof of Theorem 5.4(c). Use the notation of the proof of Theorem 3.7. We claim that for every i

$$\mathbb{E}_{\nu_i}\big[G_i\,\mathbf{1}[T \neq \emptyset]\big] \le (8\ln m + 9.86)\,\nu_i(T \neq \emptyset).$$

If $\nu_i(T \neq \emptyset) \ge \frac{1}{2}$, this follows from $\mathbb{E}_{\nu_i} G_i < 4\ln m + 4.93$. Otherwise $\nu_i(\emptyset) > \frac{1}{2}$, and since $\nu_i(\emptyset) = \widehat{g}(\{i\})^2/\mathrm{Inf}_i(f) \le \mathrm{Inf}_i(f)$ we get $\mathrm{Inf}_i(f) > \frac{1}{2}$. As $\mathrm{Inf}_i(f) \le \sum_l \mathrm{Inf}_i(h_l)$, some h_l has $\mathrm{Inf}_i(h_l) > 1/(2m)$. The influences of the regular function h_l do not increase along its order (see the proof of part (b)) and their squares sum to at most 1, so $r_l(i) < 4m^2$, and then $G_i \le F_l \le \ln r_l(i) < 2\ln(2m)$ pointwise. This proves the claim, and summing it with weights $\mathrm{Inf}_i(f)$ gives

$$\sum_{|A|\ge 2} P(A) \sum_{i\in A} \min_l \ln^+ \frac{r_l(i)}{|A|} \le (8\ln m + 9.86)\,J(f).$$

Now take $Q(A) = 2^{-(|A|-1)}\pi_{|A|}(A)$ with the weights θ_k of the proof of Theorem 3.7; as in part (b), Proposition 5.3 gives

$$H^+(f) \le \Big(\frac{2}{\ln 2} + 9 + \log\frac{ema}{a-1} + \frac{a(8\ln m + 9.86)}{\ln 2}\Big) J(f),$$

and $a = 1.2$ gives the stated bound. □

Proof of Theorem 5.4(d). By complementation it suffices to treat unions. Take $Q = Q_1$ and the constants of the proof of Theorem 4.1. The bound on $\log(1/Q_1(A))$ obtained there holds for every $A \neq \emptyset$, so

$$\sum_{|A|\ge 2} P(A)\log\frac{1}{Q_1(A)} \le W(f)\Lambda + (\beta + c_1)k\,J(f) + J(f) + \sum_j \mathrm{Inf}_j(f)\,\mathbb{E}_{\nu_j}\big[G_j\,\mathbf{1}[T \neq \emptyset]\big].$$

The vacuum term is treated as in part (a), with Lemma 4.2 in place of Lemma 2.1: $W(f)\Lambda \le 4k\,J(f)$. We claim that $\mathbb{E}_{\nu_j}[G_j \mathbf{1}[T \neq \emptyset]] \le 2k\big(\frac{aL_1}{\ln 2} + c_2K\big)\nu_j(T \neq \emptyset)$. If $\nu_j(T \neq \emptyset) \ge \frac{1}{2}$, this follows from the bound on $\mathbb{E}_{\nu_j} G_j$ in the proof of Theorem 4.1. Otherwise $\mathrm{Inf}_j(f) \ge \nu_j(\emptyset) > \frac{1}{2}$ as in part (c); since $\mathrm{Inf}_j(f) \le \sum_s \mathrm{Inf}_j(h_{l_s})$ with at most k summands, some s has $\mathrm{Inf}_j(h_{l_s}) > 1/(2k)$. Then $r_{l_s}(j) < 4k^2$ as in part (c), and $\mathbb{P}[h_{l_s} = 1] \ge \mathrm{Inf}_j(h_{l_s})/2 > 1/(4k)$, so pointwise $G_j \le G_j(l_s) < \frac{a}{\ln 2}\ln(4k^2) + c_2\log(4k) \le 2k\big(\frac{aL_1}{\ln 2} + c_2K\big)$. Summing the claim with weights $\mathrm{Inf}_j(f)$ and using Proposition 5.3,

$$H^+(f) \le \Big(\frac{2}{\ln 2} + 9 + \big(4 + \beta + c_1 + 2\,\tfrac{aL_1}{\ln 2} + 2c_2K\big)k\Big) J(f),$$

and the coefficient of k is $4 + 5 + 4.273 + 17.248 + 2.457 < 33$.

For the logarithmic bound take $Q = Q_1$ and the constants of the proof of Theorem 4.4. The bound on $\log(1/Q_1(A))$ obtained there holds for every $A \neq \emptyset$ and has no term independent of

$|A|$ other than Λ_λ, so

$$\sum_{|A|\ge 2} P(A)\log\frac{1}{Q_1(A)} \le W(f)\Lambda_\lambda + K_u\,J(f) + \sum_j \mathrm{Inf}_j(f)\,\mathbb{E}_{\nu_j}\big[G_j\,\mathbf{1}[T\neq\emptyset]\big],$$

with $K_u = (1+a)\log k+\beta+1+2\log 3+\log\frac{ea}{a-1}+\log(2e)$. As in part (a), with $\Lambda_\lambda \le (\log e)\sum_l \pi_l/k$ and Lemma 4.2, $W(f)\Lambda_\lambda \le 2(\log e)J(f)$. If $\nu_j(T\neq\emptyset)\ge\frac12$, then $\mathbb{E}_{\nu_j}[G_j\mathbf{1}[T\neq\emptyset]] \le 2E_k\,\nu_j(T\neq\emptyset)$, where $E_k = \frac{4a}{\ln 2}(\ln(2kB)+1)+\frac{2}{\gamma}(\log k+\log e)$ is the bound on $\mathbb{E}_{\nu_j}G_j$ from that proof; otherwise, as above, $G_j < a\log(4k^2)+\log(4k)$ pointwise. By Proposition 5.3,

$$H^+(f) \le \Big(\frac{2}{\ln 2}+8+2\log e+K_u+\max\big(2E_k,\,a\log(4k^2)+\log(4k)\big)\Big)J(f) \le (93+23.1\log k)\,J(f).$$

□

We recall the biased form of (1) [21]. Let μ be a product probability measure on $\{0,1\}^m$ with $q_j=\mu(y_j=1)\in(0,1)$, put $\nu_j=2q_j-1$, $\sigma_j^2=1-\nu_j^2=4q_j(1-q_j)$ and $\phi_j(y)=(2y_j-1-\nu_j)/\sigma_j$, so that the products $\phi_A=\prod_{j\in A}\phi_j$ form an orthonormal basis of $L^2(\mu)$. For $G:\{0,1\}^m\to\{0,1\}$ expand $1-2G=\sum_A\widetilde{g}(A)\phi_A$ and put $P_\mu(A)=\widetilde{g}(A)^2$, $\sigma_A^2=\prod_{j\in A}\sigma_j^2$, $I_\mu(G)=\sum_A|A|P_\mu(A)$ and $\mathrm{Var}_\mu(G)=\sum_{A\neq\emptyset}P_\mu(A)$. Then G satisfies the μ-biased form of (1) with constant C if

$$\sum_{A\neq\emptyset} P_\mu(A)\log\frac{\sigma_A^2}{P_\mu(A)} \le C\big(I_\mu(G)-\mathrm{Var}_\mu(G)\big). \tag{6}$$

For the uniform measure this is (1). Negating an input or the output of G leaves every $P_\mu(A)$ unchanged, up to replacing q_j by $1-q_j$.

Lemma 5.5. *For every m and every product measure μ on $\{0,1\}^m$, the functions OR_m and AND_m satisfy (6) with $C=2+4\log e=7.7707\ldots$, and XOR_m satisfies (6) with $C=2\log e$. The constant $2+4\log e$ cannot be lowered: for OR_2 with $q_1=q_2\to 0$ the ratio of the two sides of (6) tends to it.*

Proof. First let $a_1,\ldots,a_m\in(0,1]$, $Z=\prod_j a_j$, and

$$\Phi(a)=\sum_j (a_j-Z)\ln\frac{1}{a_j},\qquad R(a)=\sum_j(1-a_j)-1+Z.$$

We claim $\Phi\le 2R$. For $m=1$ both sides vanish. Appending a number $b\in(0,1]$ replaces Z by bZ and gives

$$\Phi(a,b)-\Phi(a)=(1-b)\,Z\ln\frac{1}{Z}+b\ln\frac{1}{b}\,(1-Z),\qquad R(a,b)-R(a)=(1-b)(1-Z),$$

and since $t\ln(1/t)\le 1-t$ for $0<t\le 1$, the first difference is at most twice the second. The claim follows by induction on m.

For OR_m it suffices, after negating the output, to take $G(y)=\prod_j(1-y_j)$, the indicator that all inputs vanish. Put $a_j=1-q_j$. Since $1-y_j=a_j-\frac{\sigma_j}{2}\phi_j$,

$$1-2G=1-2\prod_j\Big(a_j-\frac{\sigma_j}{2}\phi_j\Big),\qquad P_\mu(A)=4\prod_{j\in A}\frac{\sigma_j^2}{4}\prod_{j\notin A}a_j^2\quad(A\neq\emptyset),$$

and $P_\mu(\emptyset)=(1-2Z)^2$. Hence $\log(\sigma_A^2/P_\mu(A))=2|A|-2+2\sum_{j\notin A}\log(1/a_j)$ for $A\neq\emptyset$. Using $\frac{\sigma_j^2}{4}+a_j^2=a_j$ one finds $\sum_{A\neq\emptyset,\,A\not\ni j}P_\mu(A)=4a_jZ-4Z^2$, $I_\mu(G)=\sum_j\sigma_j^2\prod_{i\neq j}a_i=4Z\sum_j(1-a_j)$

and $\mathrm{Var}_\mu(G) = 4Z(1-Z)$, so that $I_\mu(G) - \mathrm{Var}_\mu(G) = 4ZR$ and the left side of (6) equals

$$2\big(I_\mu(G) - \mathrm{Var}_\mu(G)\big) + 8Z\sum_j (a_j - Z)\log\frac{1}{a_j} = 8ZR + \frac{8Z\,\Phi}{\ln 2} \le \Big(2 + \frac{4}{\ln 2}\Big)\big(I_\mu(G) - \mathrm{Var}_\mu(G)\big).$$

AND_m is obtained from OR_m by negating the inputs and the output. For $m = 2$ and $q_1 = q_2 = q$ we have $\Phi = 2q(1-q)\ln\frac{1}{1-q} \sim 2q^2$ and $R = q^2$, so the ratio of the two sides of (6) is $2 + 2\Phi/(R\ln 2) \to 2 + 4\log e$ as $q \to 0$.

For XOR_m, $1 - 2G = \pm\prod_j(\nu_j + \sigma_j\phi_j)$, so $P_\mu(A) = \sigma_A^2\prod_{j\notin A}\nu_j^2$ and $\log(\sigma_A^2/P_\mu(A)) = \sum_{j\notin A}\log(1/\nu_j^2)$. By continuity we may assume $\nu_j \neq 0$. With $a_j = \nu_j^2$, $\sum_{A\not\ni j} P_\mu(A) = a_j$ and $P_\mu(\emptyset) = Z$, so the left side of (6) is $\Phi/\ln 2$, while $I_\mu(G) - \mathrm{Var}_\mu(G) = \sum_j \sigma_j^2 - (1-Z) = R$. □

Threshold gates with inputs of equal bias. For $1 \le t \le r$ let $T_{t,r}(y) = \mathbf{1}[y_1 + \cdots + y_r \ge t]$ on $\{0,1\}^r$, and let μ be the product measure with $\mu(y_j = 1) = q$ for every j, where $0 < q < 1$. Write $N(y) = y_1 + \cdots + y_r$, $N_{-j} = N - y_j$, $\sigma^2 = 4q(1-q)$ and $\Lambda = r\sigma^2$, and let $P = P_\mu$, $I = I_\mu(T_{t,r})$, $V = \mathrm{Var}_\mu(T_{t,r})$ and $W_k = \sum_{|A|=k} P(A)$. Every coordinate is pivotal with the same probability $\rho = \mu(N_{-j} = t-1)$, and $I = \Lambda\rho$.

Theorem 5.6. *In this setting*

$$\sum_{A\neq\emptyset} P(A)\ln\frac{\sigma_A^2}{P(A)} \le 11.49\, I.$$

Consequently $T_{t,r}$ satisfies (6) under μ with $C = 66.4$, for every r, every $1 \le t \le r$ and every q.

The proof rests on two lemmas. The first is the analogue of Lemma 3.4.

Lemma 5.7. $\sum_{k\ge1} kW_k \ln^+(\Lambda/k) \le L' I$, *where* $L' = 2\ln\big(3\sqrt{6/\pi}\big) + 2 < 4.85$.

Proof. Let $d(y') = T_{t,r}(y',1) - T_{t,r}(y',0) = \mathbf{1}[N(y') = t-1]$ for $y' \in \{0,1\}^{r-1}$, and let $\widetilde{d}(B)$ be its coefficients in the basis ϕ_B of $L^2(\mu)$ on $\{0,1\}^{r-1}$. Since $\mathbb{E}_{y_r}[h\,\phi_r] = \frac{\sigma}{2}(h(1) - h(0))$ for every function h of y_r, we have $P(B\cup\{r\}) = \sigma^2\widetilde{d}(B)^2$, and by symmetry $\sum_{|A|=k,\,A\ni r} P(A) = \frac{k}{r}W_k$. Hence $kW_k = \Lambda\sum_{|B|=k-1}\widetilde{d}(B)^2$, and $\nu(B) = \widetilde{d}(B)^2/\rho$ is a probability distribution because $\mathbb{E}d^2 = \rho$.

Let y' have law μ, fix $0 < \eta \le 1$, and let z be obtained from y' by replacing each coordinate, independently with probability η, by a fresh sample from μ. Then $\mathbb{E}[\phi_B(z) \mid y'] = (1-\eta)^{|B|}\phi_B(y')$, so $\mathbb{E}[d(y')d(z)] = \rho\,\mathbb{E}_\nu(1-\eta)^{|T|}$, while $\mathbb{E}[d(y')d(z)] \le \rho\max_{y'}\mathbb{P}[N(z) = t-1 \mid y']$. Write each replaced coordinate as $z_j = u_j \wedge b_j$ if $q \le \frac12$ and as $z_j = u_j \vee b_j$ if $q > \frac12$, with u_j a uniform bit and b_j an independent bit equal to 1 with probability $2q$, respectively $2q-1$. Given the set of replaced coordinates and the bits b_j, the replaced coordinates with $b_j = 1$ (respectively $b_j = 0$) are independent uniform bits, say F of them, and every other coordinate of z is fixed; so by Lemma 3.2 the conditional probability that $N(z) = t-1$ is at most $\frac{2}{\sqrt{\pi}}(F+1)^{-1/2}$. The number F has the law $\mathrm{Bin}(r-1, 2\eta\tilde{q})$ with $\tilde{q} = \min(q, 1-q) \ge q(1-q)$, so $\mathbb{E}[1/(F+1)] \le 1/(2r\eta\tilde{q}) \le 2/(\eta\Lambda)$ and, by Jensen's inequality, $\mathbb{E}[(F+1)^{-1/2}] \le (2/(\eta\Lambda))^{1/2}$. For $K \ge 1$ take $\eta = 1/(3K)$; as in the proof of Lemma 3.4,

$$\nu(|T| \le K) \le \Big(1 - \frac{1}{3K}\Big)^{-K}\mathbb{E}_\nu(1-\eta)^{|T|} \le 3\sqrt{6/\pi}\,\Big(\frac{K}{\Lambda}\Big)^{1/2}.$$

Finally $\sum_k kW_k\ln^+(\Lambda/k) = I\,\mathbb{E}_\nu\ln^+\frac{\Lambda}{|T|+1}$, and the tail bound with $K = \Lambda e^{-u}$ gives

$$\mathbb{E}_\nu\ln^+\frac{\Lambda}{|T|+1} \le \int_0^\infty \min\big(1, 3\sqrt{6/\pi}\,e^{-u/2}\big)\,du = L'.$$

□

The second lemma pays for the normalisation $V\ln(1/V)$. When the inputs are nearly always 0 and $t \geq 2$ this term is much larger than I, and it is repaid by the factors σ_A^2; the lemma measures this through the ratio ρ/V.

Lemma 5.8. *$V\ln(1/V) \leq I\ln^+(4\rho/V) + C_2 I$, where $C_2 = 1 + \ln 2 + 8/e < 4.64$.*

Proof. Let $\alpha = \min(\mu(T_{t,r} = 1), \mu(T_{t,r} = 0))$. If $\alpha \geq \frac{1}{4}$ then $V \geq \frac{3}{4}$ and $V\ln(1/V) \leq V\ln\frac{4}{3} < 0.29\,I$. Let $\alpha < \frac{1}{4}$. Negating all inputs and the output maps $T_{t,r}$ to $T_{r-t+1,r}$ and q to $1-q$, and changes none of V, I, ρ; so we may assume $\alpha = \mu(A)$ with $A = \{N \geq t\}$. Then $t > rq$, since otherwise $t \leq \lfloor rq \rfloor$, which is at most the median of $\mathrm{Bin}(r,q)$ [13], and $\mu(A) \geq \frac{1}{2}$. As $V = 4\alpha(1-\alpha) \leq 4\alpha$ and $4(1-\alpha) > 1$, we have $V\ln(1/V) \leq 4\alpha\ln(1/\alpha)$.

Let $\mu_A = \mu(\cdot \mid A)$ and let $\mu_A^{(j)}$ be the law of y_j under μ_A. By the chain rule for relative entropy with respect to a product measure,

$$\ln\frac{1}{\alpha} = D(\mu_A \,\|\, \mu) = \mathrm{TC}(\mu_A) + \sum_j D\big(\mu_A^{(j)} \,\big\|\, \mu_j\big), \qquad \mathrm{TC}(\mu_A) = D\Big(\mu_A \,\Big\|\, \textstyle\bigotimes_j \mu_A^{(j)}\Big).$$

For a coordinate j put $a = \mu(A \mid y_j = 1)$ and $b = \mu(A \mid y_j = 0)$, so that $\alpha = qa + (1-q)b$ and $a - b = \rho$. Since $b \leq \alpha$, $\alpha D(\mu_A^{(j)} \| \mu_j) = qa\ln\frac{a}{\alpha} + (1-q)b\ln\frac{b}{\alpha} \leq qa\ln\frac{a}{\alpha}$. With $u = (1-q)\rho/\alpha$ we have $a/\alpha = 1+u$ and $(1+u)\ln(1+u) \leq u(1+\ln(1+u))$, hence

$$\alpha D\big(\mu_A^{(j)} \,\big\|\, \mu_j\big) \leq q(1-q)\rho\,(1+\ln(1+u)) \leq \frac{\sigma^2\rho}{4}\Big(1 + \ln 2 + \ln^+\frac{4\rho}{V}\Big),$$

using $\ln(1+u) \leq \ln 2 + \ln^+ u$ and $\alpha \geq V/4$. Summing over j gives $4\alpha\sum_j D(\mu_A^{(j)}\|\mu_j) \leq I(1 + \ln 2 + \ln^+(4\rho/V))$.

It remains to show $\alpha\,\mathrm{TC}(\mu_A) \leq \frac{2}{e} I$. If $t = r$ then μ_A is a point mass and $\mathrm{TC}(\mu_A) = 0$. Let $rq < t \leq r-1$ and let R be the product measure with $R(y_j = 1) = t/r$; its density is $dR/d\mu = M^{-1}e^{\lambda N}$ with $\lambda = \ln\frac{t(1-q)}{(r-t)q} > 0$ and $M = \mathbb{E}_\mu e^{\lambda N}$. For every product measure R we have $D(\mu_A\|R) = \mathrm{TC}(\mu_A) + \sum_j D(\mu_A^{(j)}\|R_j) \geq \mathrm{TC}(\mu_A)$, and here

$$D(\mu_A \,\|\, R) = \ln\frac{M}{\alpha} - \lambda\,\mathbb{E}_{\mu_A} N \leq \ln\frac{Me^{-\lambda t}}{\alpha}, \qquad \alpha = M\,\mathbb{E}_R\big[e^{-\lambda N}\mathbf{1}_A\big] \geq Me^{-\lambda t}R(N = t).$$

Hence $\mathrm{TC}(\mu_A) \leq \ln(1/b^*)$, where $b^* = R(N=t) = \binom{r}{t}(t/r)^t(1-t/r)^{r-t} \geq (8v)^{-1/2}$ with $v = t(r-t)/r \geq \frac{1}{2}$ [18, Ch. 10, Lemma 7]. In the same way $\rho = Me^{-\lambda(t-1)}R(N_{-j} = t-1)/(1-q+qe^\lambda)$ and $\alpha \leq Me^{-\lambda t}$, and $R(N_{-j} = t-1) = \binom{r-1}{t-1}(t/r)^{t-1}(1-t/r)^{r-t} = b^*$; since $\sigma^2 e^\lambda/(1-q+qe^\lambda) = 4\frac{t}{r}(1-\frac{t}{r})(1-q+qe^\lambda) \geq 4\frac{t}{r}(1-\frac{t}{r})$, this gives $I/\alpha \geq 4vb^* \geq (2v)^{1/2}$. Therefore

$$\alpha\,\mathrm{TC}(\mu_A) \leq \frac{1}{2}\ln(8v)\,\frac{I}{(2v)^{1/2}} = \frac{\ln(8v)}{\sqrt{8v}}\,I \leq \frac{2}{e}\,I,$$

and altogether $V\ln(1/V) \leq 4\alpha\ln(1/\alpha) \leq I\ln^+(4\rho/V) + (1 + \ln 2 + 8/e)I$. □

Proof of Theorem 5.6. By symmetry $P(A) = W_k/\binom{r}{k}$ when $|A| = k$, so the left side equals

$$\sum_{k\geq 1} W_k \ln\frac{1}{W_k} + \sum_{k\geq 1} W_k \ln\Big(\binom{r}{k}\sigma^{2k}\Big).$$

The first sum is $V\ln(1/V) + V\,H(K)$, where K has law W_k/V on $\{1,\dots,r\}$ and mean $m = I/V$. Gibbs' inequality against the geometric law with mean m gives $H(K) \leq 1 + \ln m$, and $V(1+\ln(I/V)) \leq I$ because $x(1+\ln(1/x)) \leq 1$ for $0 < x \leq 1$. For the second sum, $\binom{r}{k} \leq$

$(er/k)^k$ gives $\ln(\binom{r}{k}\sigma^{2k}) \le k + k\ln(\Lambda/k)$. If $4m \le \Lambda$, then $\ln^+(4\rho/V) = \ln^+(4m/\Lambda) = 0$ and $\sum_k kW_k \ln(\Lambda/k) \le L'I$ by Lemma 5.7. If $4m > \Lambda$, write $\ln\frac{\Lambda}{k} = -\ln\frac{4m}{\Lambda} + \ln\frac{4m}{k}$ and use $\ln x \le x/e$: $\sum_k kW_k \ln(\Lambda/k) \le -I\ln(4\rho/V) + \frac{4m}{e}V$, and $4mV/e = 4I/e < L'I$. In both cases the left side is at most

$$V\ln\frac{1}{V} + 2I - I\ln^+\frac{4\rho}{V} + L'I \le (2 + L' + C_2)\,I < 11.49\,I$$

by Lemma 5.8.

For the second statement, Lemma 5.5 covers $t \in \{1, r\}$. Let $2 \le t \le r-1$. The number $\rho = \mathbb{P}[\mathrm{Bin}(r-1,q) = t-1]$ is the value of a binomial distribution at an interior point; it is $2q(1-q) \le \frac{1}{2}$ when $r = 3$, and for $r \ge 4$ it is at most $\max_{1\le i\le m-1}\binom{m}{i}(i/m)^i(1-i/m)^{m-i} \le (m/(2\pi(m-1)))^{1/2} < \frac{1}{2}$ with $m = r-1$ [18]. Since $d \ge 0$ we have $P(\{j\}) = \sigma^2(\mathbb{E}d)^2 = \sigma^2\rho^2$ and $\sum_{A\ni j} P(A) = \sigma^2\rho$, so $\sum_{|A|\ge 2}|A|P(A) = I - \Lambda\rho^2 \ge I/2$ and $I - V = \sum_{|A|\ge 2}(|A|-1)P(A) \ge I/4$. In bits the left side of (6) is therefore at most $\frac{4\cdot 11.49}{\ln 2}(I - V) < 66.4\,(I - V)$. □

Remark 5.9. The same reduction shows where general Boolean functions behave differently. For the disjunction of s disjoint conjunctions of two inputs, each input equal to 1 with probability s^{-2}, the total correlation of μ_A grows like $\ln s$ (it records which conjunction is satisfied) while I_μ/α stays bounded, and the inequality of Lemma 5.8, with $\sum_j \mathrm{Inf}_j \ln^+(4\rho_j/V)$ in place of $I\ln^+(4\rho/V)$, fails for these functions.

Threshold gates under arbitrary product measures. Under an arbitrary product measure the same argument proves the inequality with I_μ in place of $I_\mu - \mathrm{Var}_\mu$ on the right. For read-once formulas this weaker form costs a factor of the depth.

Proposition 5.10. *Let $1 \le t \le r$ and let μ be a product measure on $\{0,1\}^r$ with $\mu(y_j = 1) = q_j \in (0,1)$. Then*

$$\sum_{A\ne\emptyset} P_\mu(A)\ln\frac{\sigma_A^2}{P_\mu(A)} \le 26.3\, I_\mu(T_{t,r}).$$

Proof. Write N, N_{-j} and W_k as before, $\sigma_j^2 = 4q_j(1-q_j)$, $\Lambda = \sum_j \sigma_j^2$, $\rho_j = \mu(N_{-j} = t-1)$, $\mathrm{Inf}_j = \sigma_j^2\rho_j$, $I = \sum_j \mathrm{Inf}_j$, $V = \mathrm{Var}_\mu(T_{t,r})$, $m = I/V$ and $\pi_j = \mathrm{Inf}_j/V$, so that $\sum_j \pi_j = m$.

For $k \ge 1$ let $\theta_k(j) = \max\bigl(\sigma_j^2\min(1, k/\Lambda),\, 4\pi_j\bigr)$ and let Z_k be the k-th elementary symmetric polynomial of $\theta_k(1),\dots,\theta_k(r)$. Gibbs' inequality on the sets of size k, against the weights $\prod_{j\in A}\theta_k(j)$, gives

$$\sum_{|A|=k} P(A)\ln\frac{\sigma_A^2}{P(A)} \le W_k\ln\frac{1}{W_k} + W_k\ln Z_k + \sum_{|A|=k} P(A)\sum_{j\in A}\ln\frac{\sigma_j^2}{\theta_k(j)}.$$

Since $\sum_j \theta_k(j) \le \min(\Lambda, k) + 4m \le k + 4m$, we have $Z_k \le (e(k+4m)/k)^k$ and $\ln Z_k \le k + 4m$. Moreover $\ln(\sigma_j^2/\theta_k(j)) \le \min\bigl(\ln^+(\Lambda/k), \ln(V/(4\rho_j))\bigr) \le \ln^+(\Lambda/k) - \ln^+(4\rho_j/V)$. Summing over k, and bounding $\sum_k W_k\ln(1/W_k)$ by $V\ln(1/V) + I$ as in the proof of Theorem 5.6,

$$\sum_{A\ne\emptyset} P(A)\ln\frac{\sigma_A^2}{P(A)} \le V\ln\frac{1}{V} + 6I + \sum_j I'_j - \sum_j \mathrm{Inf}_j\ln^+\frac{4\rho_j}{V}, \qquad I'_j = \sum_{A\ni j} P(A)\ln^+\frac{\Lambda}{|A|}.$$

For the rank terms, $I'_j = \mathrm{Inf}_j\,\mathbb{E}_{\nu_j}\ln^+\frac{\Lambda}{|T|+1}$ with $\nu_j(T) = P(T\cup\{j\})/\mathrm{Inf}_j$. If $\Lambda \le 2$ the expectation is at most $\ln 2$. If $\Lambda > 2$, the proof of Lemma 5.7 applies to the coordinate j, except that a replaced coordinate i is now free with probability $2\eta\tilde{q}_i \ge \eta\sigma_i^2/2$, so that F is a sum of independent indicators with mean at least $\eta(\Lambda - \sigma_j^2)/2 \ge \eta\Lambda/4$; for such a sum $\mathbb{E}[1/(F+1)] \le$

$1/\mathbb{E}F$, hence $\nu_j(|T|\le K)\le 3\sqrt{12/\pi}\,(K/\Lambda)^{1/2}$ and $\mathbb{E}_{\nu_j}\ln^+\frac{\Lambda}{|T|+1}\le 2\ln\!\big(3\sqrt{12/\pi}\big)+2<5.54$. So $\sum_j I'_j\le 5.54\,I$.

It remains to show $V\ln(1/V)\le\sum_j \mathrm{Inf}_j\ln^+(4\rho_j/V)+14.7\,I$, which gives the constant $6+5.54+14.7<26.3$. We follow the proof of Lemma 5.8. If $\alpha\ge\frac14$ the bound is immediate. Otherwise, after negating all inputs if necessary, $\alpha=\mu(A)$ with $A=\{N\ge t\}$, and $t>\mathbb{E}_\mu N$: if $t\le\mathbb{E}_\mu N=r\bar q$, then $t\le\lfloor r\bar q\rfloor$, and Hoeffding's comparison with the binomial law [11, Theorem 4] together with [13] gives $\mu(A)\ge\mathbb{P}[\mathrm{Bin}(r,\bar q)\ge t]\ge\frac12$. The bound $4\alpha D(\mu_A^{(j)}\|\mu_j)\le\mathrm{Inf}_j(1+\ln 2+\ln^+(4\rho_j/V))$ holds for each coordinate as before. For the total correlation, if $t=r$ it vanishes; otherwise let R be the product measure with $dR/d\mu\propto e^{\lambda N}$, with $\lambda>0$ chosen so that $\mathbb{E}_R N=t$. As in the proof of Lemma 5.8, $\mathrm{TC}(\mu_A)\le\ln(1/R(N=t))$ and, for each j, $\sigma_j^2\rho_j/\alpha\ge 4R(y_j{=}1)R(y_j{=}0)R(N_{-j}=t-1)$. Applying the identity $\mathbb{E}_R[(y_j-R(y_j{=}1))h(N)]=R(y_j{=}1)R(y_j{=}0)\,\mathbb{E}_R[h(N_{-j}+1)-h(N_{-j})]$ to $h=\mathbf{1}[\,\cdot\ge t]$ and summing over j gives $\sum_j R(y_j{=}1)R(y_j{=}0)R(N_{-j}=t-1)=\mathbb{E}_R(N-t)^+$, so $I/\alpha\ge 4\mathbb{E}_R(N-t)^+=2\mathbb{E}_R|N-t|$. With $v=\mathrm{Var}_R(N)$ we have $\mathbb{E}_R(N-t)^4\le v+3v^2$, and Hölder's inequality gives $\mathbb{E}_R|N-t|\ge v^{3/2}(v+3v^2)^{-1/2}$; hence $I/\alpha\ge 2v(1+3v)^{-1/2}$. By Darroch's theorem [5], t is a mode of N under R. If $v\le\frac12$, Chebyshev's inequality gives $R(N=t)\ge 1-v$ and $\mathrm{TC}(\mu_A)\le\ln\frac{1}{1-v}\le 2v\ln 2$, so $\alpha\,\mathrm{TC}(\mu_A)\le(\ln 2)(1+3v)^{1/2}I<1.1\,I$. If $v>\frac12$, let $K=\lceil 2\sqrt{v}\rceil$; by Chebyshev's inequality $R(|N-t|\le K)\ge\frac34$, so $R(N=t)\ge\frac{3}{4(2K+1)}\ge\frac{3}{4(4\sqrt{v}+3)}$ and $\alpha\,\mathrm{TC}(\mu_A)\le\psi(v)I$ with $\psi(v)=\ln\frac{16\sqrt{v}+12}{3}\cdot\frac{(1+3v)^{1/2}}{2v}$. Comparing logarithmic derivatives shows that ψ decreases on $[\frac12,\infty)$, and $\psi(\frac12)<3.25$. Hence $\alpha\,\mathrm{TC}(\mu_A)\le 3.25\,I$ and $V\ln(1/V)\le 4\alpha\ln(1/\alpha)\le\sum_j\mathrm{Inf}_j\ln^+(4\rho_j/V)+(1+\ln 2+13)I$. □

Lemma 5.11. *Let $F=G(f_1,\dots,f_k)$, where $f_1,\dots,f_k$ depend on pairwise disjoint sets of variables, and let μ be the product measure with $\mu(y_i=1)=\mathbb{P}[f_i=1]$. If G satisfies (6) under μ with $I_\mu(G)$ in place of $I_\mu(G)-\mathrm{Var}_\mu(G)$ and with constant C, and $H^+(f_i)\le D\,I(f_i)$ for every i, then $H^+(F)\le(C+D)\,I(F)$.*

Proof. By the composition formula of [21, Section 3], the spectral distribution of F is obtained by choosing A with law P_μ and then, independently for $i\in A$, a nonempty set T_i of variables of f_i with law $\widehat{f_i}(T_i)^2/\sigma_i^2$, where $\sigma_i^2=\mathrm{Var}(f_i)$. Since $\sum_{T\ne\emptyset}\widehat{f_i}(T)^2=\sigma_i^2$, this gives

$$H^+(F)=\sum_{A\ne\emptyset}P_\mu(A)\log\frac{\sigma_A^2}{P_\mu(A)}+\sum_i\frac{\mathrm{Inf}_i^\mu(G)}{\sigma_i^2}H^+(f_i),\qquad I(F)=\sum_i\frac{\mathrm{Inf}_i^\mu(G)}{\sigma_i^2}I(f_i),$$

and the claim follows from $\mathrm{Inf}_i^\mu(G)=\frac{\mathrm{Inf}_i^\mu(G)}{\sigma_i^2}\mathrm{Var}(f_i)\le\frac{\mathrm{Inf}_i^\mu(G)}{\sigma_i^2}I(f_i)$. □

Corollary 5.12. *Let f be computed by a read-once formula of depth d whose gates are* AND, OR *and* XOR *gates and threshold gates $T_{t,r}$ of any fan-in, with negations allowed. Then $H(f)\le(38d+\log e)\,I(f)$.*

Proof. By Proposition 5.10 and Lemma 5.5, every gate satisfies the hypothesis of Lemma 5.11 under every product measure with $C=38>26.3/\ln 2$; negating an input replaces q_j by $1-q_j$ and changes nothing else. A literal has $H^+=0$. By induction on the depth, Lemma 5.11 gives $H^+(f)\le 38d\,I(f)$, and $H(f)=H^+(f)+(1-\mathrm{Var}(f))\log\frac{1}{1-\mathrm{Var}(f)}\le H^+(f)+\mathrm{Var}(f)\log e\le(38d+\log e)\,I(f)$. □

The inputs of such a formula may also be functions from the classes of Theorem 1.3, on disjoint sets of variables, at the cost of adding their constant. For formulas of bounded depth the constant

does not depend on the fan-ins, while [21, Theorem 3] gives a constant independent of the depth that grows exponentially with the fan-ins. The next theorem removes the dependence on d.

The strengthened inequality for threshold gates. The weak form loses only when one input is pivotal with probability close to one. The other inputs are then nearly deterministic, and the left side of (6) can be estimated directly.

Theorem 5.13. *Let $1 \le t \le r$ and let μ be a product measure on $\{0,1\}^r$ with $\mu(y_j = 1) = q_j \in (0,1)$. With $J = \sum_{|A|\ge 2} |A|\, P_\mu(A)$,*

$$\sum_{A \neq \emptyset} P_\mu(A) \ln \frac{\sigma_A^2}{P_\mu(A)} \le 263\, J \le 526 \left(I_\mu(T_{t,r}) - \mathrm{Var}_\mu(T_{t,r}) \right),$$

so $T_{t,r}$ satisfies (6) with $C = 760$ under every product measure.

Proof. Write N, N_{-j}, σ_j^2, $\rho_j = \mu(N_{-j} = t-1)$, I and V as in Proposition 5.10, and put $J_j = \sigma_j^2 \rho_j (1-\rho_j)$. The derivative $d_j = \mathbf{1}[N_{-j} = t-1]$ has mean ρ_j and $P(B \cup \{j\}) = \sigma_j^2 \widetilde{d}_j(B)^2$ (proof of Lemma 5.7), so $P(\{j\}) = \sigma_j^2 \rho_j^2$, $\sum_{A \ni j,\, |A|\ge 2} P(A) = \sigma_j^2 \mathrm{Var}(d_j) = J_j$ and $J = \sum_j J_j$. Since $J \ge 2 \sum_{|A|\ge 2} P(A)$, we have $I - V = \sum_{|A|\ge 2} (|A|-1) P(A) \ge J/2$, which is the second inequality. The sets of size one contribute $\sum_j 2\sigma_j^2 \rho_j^2 \ln(1/\rho_j) \le 2J$.

If $\rho_j \le \frac{9}{10}$ for every j, then $J \ge I/10$ and Proposition 5.10 gives the bound $263\, J$. Otherwise let j_0 maximise ρ_j, so that $\rho_{j_0} = 1 - \varepsilon$ with $\varepsilon < \frac{1}{10}$. Negating all inputs and the output maps $T_{t,r}$ to $T_{r-t+1,r}$ and changes none of the quantities involved, so we may assume $q := q_{j_0} \le \frac{1}{2}$. Let $O = [r] \setminus \{j_0\}$ and $N' = N_{-j_0}$. For $i \in O$ let $\tilde{q}_i = \min(q_i, 1-q_i)$, let $\bar{y}_i$ be the more likely value of y_i and $z_i = \mathbf{1}[y_i \neq \bar{y}_i]$; let $U = \{i \in O : \bar{y}_i = 0\}$, $D = \{i \in O : \bar{y}_i = 1\}$, $Z_U = \sum_{i\in U} z_i$, $Z_D = \sum_{i \in D} z_i$, $L_U = \sum_{i\in U} \tilde{q}_i$, $L_D = \sum_{i \in D} \tilde{q}_i$, $L' = L_U + L_D$, $\Lambda_U = \sum_{i\in U} \sigma_i^2$, $\Lambda_D = \sum_{i\in D} \sigma_i^2$ and $\Lambda' = \Lambda_U + \Lambda_D$.

The other inputs are nearly deterministic. Resampling every coordinate, as in the proof of Lemma 5.7 with $\eta = 1$, and using $\mathbb{E}[1/(F+1)] \le 1/\mathbb{E}F$ as in the proof of Proposition 5.10, gives $\mu(N' = k) \le (2/(\pi L'))^{1/2}$ for every k, because the number F of free coordinates has mean $2L'$. If $|D| \neq t-1$, then $N' = t-1$ forces some $z_i = 1$, so $\prod_{i\in O}(1-\tilde{q}_i) \le \varepsilon < \frac{1}{10}$; as $\ln(1-x) \ge -\frac{3}{2}x$ for $0 \le x \le \frac{1}{2}$, this gives $L' > \frac{2}{3}\ln 10 > 1.53$ and $\rho_{j_0} < 0.65$, which is impossible. Hence $|D| = t-1$ and $N' - (t-1) = Z_U - Z_D$. A configuration with exactly one $z_i = 1$ has $N' \neq t-1$, so $\varepsilon \ge L'(1-L')$; since also $L' \le 2/(\pi \rho_{j_0}^2) < 0.79$, this forces $L' < 0.113$, and then $L' \le 1.128\,\varepsilon$ and $\Lambda' \le 4L' \le 4.51\,\varepsilon \le 0.451$.

Lower bounds for J. For $i \in U$, $N_{-i} = t-1$ exactly when $y_{j_0} + Z_{U\setminus\{i\}} - Z_D = 0$, so $\rho_i \ge (1-q)(1-L') > 0.44$, while $\rho_i \le 1-\varepsilon$; by concavity $\rho_i(1-\rho_i) \ge \min(0.44 \cdot 0.56,\, \varepsilon(1-\varepsilon)) \ge 0.9\,\varepsilon$. For $i \in D$, $N_{-i} = t-1$ exactly when $y_{j_0} + Z_U - Z_{D\setminus\{i\}} = 1$, so $\rho_i \ge q(1-L') \ge 0.887\, q$ and $1 - \rho_i \ge (1-q)(1-L') > 0.44$, whence $\rho_i(1-\rho_i) \ge 0.49\, q$. Therefore $\sum_{i\in U} J_i \ge 0.9\,\varepsilon \Lambda_U$ and $\sum_{i\in D} J_i \ge 0.49\, q \Lambda_D$.

Sets containing j_0. With $d = d_{j_0}$, the sets $A \ni j_0$ with $|A| \ge 2$ contribute

$$\sigma_{j_0}^2 \sum_{B \neq \emptyset} \widetilde{d}(B)^2 \ln \frac{\sigma_B^2}{\widetilde{d}(B)^2}.$$

Let U_k be the weight of d at level k, $U_d = \mathrm{Var}(d) = \varepsilon(1-\varepsilon)$, and e_k the k-th elementary symmetric polynomial of $(\sigma_i^2)_{i\in O}$, so that $e_k \le \Lambda'^k \le (5.02\, U_d)^k$. Gibbs' inequality on each level

against the weights σ_B^2, together with $x\ln(a/x) \le a/e$, gives

$$\sum_{B\neq\emptyset} \widetilde{d}(B)^2 \ln\frac{\sigma_B^2}{\widetilde{d}(B)^2} \le \sum_k U_k \ln\frac{e_k}{U_k} \le \sum_k \Big(\frac{U_d^k}{e} + kU_k \ln 5.02\Big) \le 0.41\,U_d + 1.62\,I(d),$$

and $I(d) = \sum_{i\in O} q_i(1-q_i)\,\mu(i \text{ is pivotal for } d) \le \Lambda'/4 \le 1.26\,U_d$. So these sets contribute at most $2.5\,\sigma_{j_0}^2 U_d = 2.5\,J_{j_0}$.

Sets avoiding j_0. Put $u = \mathbf{1}[Z_U - Z_D \ge 1]$ and $v = \mathbf{1}[Z_D - Z_U \ge 1]$. Since $T_{t,r} = u + y_{j_0}(1-u-v)$, for $\emptyset \ne A \subseteq O$ the coefficient of ϕ_A in the ±1 version of $T_{t,r}$ is $2(1-q)\tilde{u}(A) - 2q\tilde{v}(A)$, where $\tilde{u}$ and $\tilde{v}$ are the coefficients of u and v. Computing in one coordinate of A as in Lemma 5.7 and using $\mathbb{E}|\phi_i| = \sigma_i$ gives $|\tilde{u}(A)|, |\tilde{v}(A)| \le \sigma_A/2$. Moreover, for $A \subseteq D$ we have $\tilde{u}(A) = \mathbb{E}\big[\mathbb{E}[u \mid y_D]\,\phi_A\big]$ with $0 \le \mathbb{E}[u \mid y_D] \le \mu(Z_U \ge 1) \le L_U$, so $|\tilde{u}(A)| \le L_U\sigma_A$, and likewise $|\tilde{v}(A)| \le L_D\sigma_A$ for $A \subseteq U$. We use three elementary facts. For $s > 0$ and $a^2, b^2 \le s$, the function $\varphi_s(x) = x\ln(s/x)$ satisfies $\varphi_s((a+b)^2) \le 2\varphi_s(a^2) + 2\varphi_s(b^2) + 2a^2 + 2b^2$: if $2a^2 + 2b^2 \le s/e$ this follows from the monotonicity of φ_s on $[0, s/e]$ and $(x+y)\ln\frac{s}{x+y} \le x\ln\frac{s}{x} + y\ln\frac{s}{y}$, and otherwise $\varphi_s \le s/e < 2a^2+2b^2$. For a family $\mathcal{F}$ of sets and numbers $0 \le x_A \le \lambda\sigma_A^2$ with $\lambda \le 1/e$, the log-sum inequality gives $\sum_{\mathcal{F}} x_A \ln(\sigma_A^2/x_A) \le \lambda\ln(1/\lambda)\sum_{\mathcal{F}}\sigma_A^2$, and $\frac{1}{e}\sum_{\mathcal{F}}\sigma_A^2$ is a bound when only $x_A \le \sigma_A^2$ is known. Finally, over the sets with $|A| \ge 2$ that meet U, $\sum \sigma_A^2 \le \Lambda_U(e^{\Lambda'} - 1) \le 1.57\,\Lambda'\Lambda_U$, and over the sets $A \subseteq D$ with $|A| \ge 2$, $\sum \sigma_A^2 \le \frac{1}{2}\Lambda_D^2 e^{\Lambda_D} \le 0.79\,\Lambda_D^2$; the same holds with U and D exchanged. We split the sets $A \subseteq O$ with $|A| \ge 2$ according to whether they meet U for the terms with $a = 2(1-q)\tilde{u}(A)$, and according to whether they meet D for the terms with $b = 2q\tilde{v}(A)$. With $\lambda = 1$, $4L_U^2$, q^2 and $4q^2L_D^2$ in the four families, $L_U \le \Lambda_U/2$, $L_D \le \Lambda_D/2$, $x^2(1 + 2\ln\frac{1}{x}) \le 0.91\,x$ for $0 < x \le 0.23$ and $q(1 + 2\ln\frac{1}{q}) \le 1.2$ for $q \le \frac{1}{2}$, these sets contribute at most

$$19.4\,\varepsilon\Lambda_U + 2.93\,\varepsilon\Lambda_U + 1.7\,q\Lambda_D + 0.3\,q\Lambda_D \le 24.9\sum_{i\in U} J_i + 4.1\sum_{i\in D} J_i.$$

Adding the three contributions, the left side is at most $2J + 2.5\,J_{j_0} + 24.9\,(J - J_{j_0}) \le 26.9\,J$. □

Corollary 5.14. *Let f be computed by a read-once formula whose gates are threshold gates $T_{t,r}$ and* XOR *gates of any fan-in, with negations allowed. Then $H(f) \le 760\,I(f)$.*

Proof. By Theorem 5.13 and Lemma 5.5, every gate satisfies (6) with $C = 760$ under every product measure, and negating an input changes nothing. The composition theorem [21, Theorem 1], applied at each gate from the inputs to the output, gives $H^+(f) \le 760\,(I(f) - \mathrm{Var}(f))$, and $H(f) \le H^+(f) + \mathrm{Var}(f)\log e \le 760\,I(f)$. □

The inputs of such a formula may again be functions from the classes of Theorem 1.3, and gates of bounded arity may be added, with the constants of Corollary 1.4. For depth $d \le 19$, Corollary 5.12 gives the smaller constant $38d + \log e$, and the numerical ratio in Section 6 is below $2 + 4\log e$ for every threshold gate that was computed.

Proof of Corollary 1.4. By [21, Theorem 2], every gate of arity at most a satisfies the biased form of (1) with a constant $C_a = 2^{O(a)}$, for every product measure on its inputs, and by Lemma 5.5 so do AND, OR and XOR gates of any fan-in, with constant 7.78; by Theorem 5.6 so does a threshold gate whose inputs have equal biases, with constant 66.4, and by Theorem 5.13 so does every threshold gate, with constant 760. By Theorem 1.3, each f_l satisfies (1) with the constant of its class, which is at least 50, and so does every gate of the last kind, because its inputs are independent balanced bits and the product measure that [21, Theorem 1] puts on them is the uniform one. The composition theorem [21, Theorem 1], applied at each gate of F from the inputs to the output, shows that f satisfies (1) with constant C, and [21, Proposition 2.1] gives the bound on $H(f)$. □

Remark 5.15. Only the uniform form of (1) is needed for the inputs f_l, because they read the uniformly distributed variables directly, and for gates with balanced inputs. Using read-k DNF formulas or regular functions as gates with biased inputs would require the biased form of (1) for these classes, which the proofs above do not give, except for AND and OR (Lemma 5.5) and for threshold gates (Theorems 5.6 and 5.13).

6. Numerical examples

All quantities below were computed exactly from full truth tables by the Walsh–Hadamard transform. For read-k DNF formulas the table lists H/I and the right side of (2), evaluated exactly with the reference measure of Theorem 2.3 and divided by I; this is the quantity that the proof of Theorem 2.3 bounds from above.

formula	n	k	H/I	Gibbs bound $/I$
tribes, 3 terms of width 5	15	1	3.632	3.632
stars of a random 3-regular graph on 12 vertices	18	2	4.128	4.366
stars of a random 3-regular graph on 16 vertices	24	2	4.487	4.724
stars of a random 4-regular graph on 12 vertices	24	2	3.875	4.004
random monotone read-3 formulas, larger of 2	20	3	3.942	4.145
random read-k formulas with negations, 192 samples	≤ 16	≤ 5	3.781	4.524

Here the star function of a graph G has one variable per edge and is true when some vertex has all its incident edges present; it is computed by a read-2 DNF formula with one term per vertex. The last row lists the largest values over the 192 formulas; the two maxima need not be attained by the same formula. On all these examples the Gibbs bound exceeds the entropy by at most $1.32\,I$, and the largest ratio is $H/I = 4.49$, for the star function of a 3-regular graph. For the reference measure of Theorem 2.5 with $\beta = 3$, the upper bound on the right side of (2) given by the runs constructed in its proof, computed from the exact spectral distribution of 28 random read-k formulas with $k \le 10$ and 12 or 14 variables, lies between $4.5\,I$ and $8.4\,I$ and grows slowly with k, while $H/I \le 3.91$ on the same formulas; the probabilities of these runs sum to at most 0.12 over all sets. For regular functions, 590 random linear threshold functions and 440 random regular functions obtained as up-sets of shifted families, all with at most 13 variables, have $H/I \le 3.94$. Intersections, unions, parities and majorities of two to four linear threshold functions on at most 12 variables, with independent random weights or with weights in opposite orders, have $H/I \le 3.86$; for these the quantity $\mathbb{E}_S \sum_{i\in S} \min_l \ln^+(r_l(i)/|S|)$ in the proof of Theorem 3.7 is at most $0.24\,I(f)$, and the expectation of F_l under each piece $\nu_{i,l}$ is at most 0.74, against the bounds $4\ln m + 4.93$ and L_1. For 75 read-k unions of threshold functions with $k \le 4$, up to 12 threshold functions and at most 14 variables, 36 of them built from rarely true threshold functions on 5 to 9 variables, $H/I \le 4.0$, the ratio of the two sides of Lemma 4.2 is at most 0.96, the expectation of $F_{l_s}^j$ under each piece is at most 0.73, the quantity bounded by K in the proof of Theorem 4.1 is at most 0.30, and the upper bound on $H(f)$ given by the choice of L_A in that proof, computed from the exact spectral distribution, lies between $H(f)$ and $37\,I(f)$. For the reference measure of Theorem 4.4 with $a = 1.44$ and $\beta = 10$, the bound given by the runs constructed in its proof, on 13 random read-k unions of threshold functions with $k \le 5$ and 10 or 12 variables, lies between $4.9\,I$ and $7.9\,I$, and the run probabilities sum to at most 0.17 over all sets. For the reference measure of Theorem 4.5, on 22 random read-k unions of intersections of two or three threshold functions with $k \le 3$ and 10 or 12 variables, the bound lies between $4.0\,I$ and $7.2\,I$, the run probabilities sum to at most 0.16, and the per-set inequality of the proof holds on every set with a margin of at least 13 bits; on 15 read-k unions of random functions of $w \le 5$ variables each ($k \le 3$, 10 or 12 variables), viewed as functions of

dictators, the bound lies between $4.0\,I$ and $7.6\,I$, the run probabilities sum to at most 0.21, and the margin is at least 14 bits.

For the strengthened inequality, the largest values of $H^{+}(f)/(I(f)-\operatorname{Var}(f))$ observed were 6.35 on random read-k DNF formulas with $k \le 6$ and at most 14 variables, 6.11 on 320 random regular and linear threshold functions with at most 12 variables, 5.68 on intersections, unions, parities and majorities of two to four linear threshold functions on the same variables, and 6.38 on conjunctions, disjunctions, parities and majorities of such functions on disjoint sets of variables. In Lemma 5.2 the largest observed value of $W(f)\log(1/W(f))/J(f)$, over all functions of at most 4 variables, random functions of up to 10 variables and perturbations of dictators, was 1.59.

For threshold gates with inputs of equal bias every quantity in Theorem 5.6 was computed with 50-digit arithmetic from the formula $W_k = \binom{r}{k}(q(1-q))^k\big(\mathbb{E}\,\Delta^k G(c)\big)^2$, where $G(c) = 1 - 2\cdot\mathbf{1}[c \ge t]$, Δ is the forward difference and c has the law $\operatorname{Bin}(r-k,q)$: for all $r \le 12$, all $1 \le t \le r$ and $q \in \{10^{-e}, 1-10^{-e} : e = 1,2,4,8,16,30\} \cup \{0.1,0.3,0.5,0.7,0.9\}$, and for 150 random triples with $13 \le r \le 40$. The left side of Theorem 5.6 divided by I is at most 2.94 (the bound is 11.49); in Lemma 5.7 the ratio is at most 1.14 (bound 4.85); in Lemma 5.8 the quantity $(V\ln(1/V) - I\ln^{+}(4\rho/V))/I$ is at most 0.68 (bound 4.64) and $\alpha\,\mathrm{TC}(\mu_A)/I$ is at most 0.25 (bound $2/e$). The ratio of the two sides of (6) is at most 7.7708, the value $2+4\log e$ of Lemma 5.5, attained by OR_2 as $q \to 0$. For linear threshold functions with at most 13 variables under 3,000 random product measures with biases down to 10^{-12}, $\alpha\,\mathrm{TC}(\mu_A) \le 0.24\,I_\mu$ as well, while for the disjunctions of Remark 5.9 the ratio grows from 0.23 to 0.39 as s goes from 3 to 7. For 3,000 further linear threshold functions with at most 12 variables, with weights in $\{1,2,3\}$, powers of two up to 2^6, or exponentially distributed, $\alpha\,\mathrm{TC}(\mu_A) \le 0.31\,I_\mu$; for these the bound $\mathrm{TC}(\mu_A) \le \ln(\chi^*/\alpha)$ given by the exponential tilt, where χ^* is the Chernoff bound for α, is far from sufficient: $\alpha\ln(\chi^*/\alpha)/I_\mu$ reaches $1.9\cdot 10^3$ on an event of probability about 10^{-16}. Over 1,500 random linear threshold functions with at most 9 variables, and 60 searches that maximise the ratio of the two sides of (6) over the weights, the threshold and the biases of a linear threshold function with at most 8 variables, that ratio never exceeded $2+4\log e$; nor did it for one heavy input and up to 32 light inputs of a common bias, computed exactly by level over a grid of biases and thresholds, where the largest value is always attained at the conjunction or disjunction corner. For 1,200 threshold functions with at most 8 variables and unequal biases down to 10^{-14}, the per-level Gibbs bound in the proof of Proposition 5.10, computed exactly, lies between the left side and $2.1\,I_\mu$ (the bound is $26.3\,I_\mu$), the rank terms are at most $0.79\,I_\mu$ and $\alpha\,\mathrm{TC}(\mu_A)$ is at most $0.17\,I_\mu$; and for 1,500 threshold functions with at most 9 variables the ratio of the two sides of (6) is again at most $2+4\log e$. For Theorem 5.13, 1,600 threshold functions with at most 10 variables were computed in 50-digit arithmetic, half of them built so that one input is pivotal with probability above $\frac{9}{10}$. In that case $L' \le 1.05\,\varepsilon$ (the proof uses $1.128\,\varepsilon$), $\Lambda' \le 4.07\,\varepsilon$ (it uses $4.51\,\varepsilon$), the sets containing j_0 contribute at most $1.57\,J_{j_0}$ (it uses $2.5\,J_{j_0}$), the sets avoiding j_0 at most 6% of the bound used for them, and the left side is at most $2.7\,J$ (it uses $26.9\,J$); in the other case the left side is at most $2.1\,I_\mu$.

7. Concluding remarks

The proofs for read-once formulas [21], for regular functions (Theorems 1.2 and 3.7) and for read-k DNF formulas (Theorem 1.1) use unrelated mechanisms: composition, a rank order with the Littlewood–Offord inequality, and activation of witnessing terms. Theorem 3.7 shows that the rank mechanism tolerates several unrelated orders at a logarithmic cost, Theorem 4.1 that the last two mechanisms combine when the regular functions overlap boundedly, and Theorem 4.5 that the three combine. The rank order suits threshold-like functions and has no analogue

for read-once formulas such as recursive majority, whose automorphism group is transitive on the variables. For the majority function each variable lies in exponentially many minterms, and the contributions of the corresponding subcubes to the derivative superpose coherently, so Lemma 2.2 gives nothing.

For read-k DNF formulas the dependence on k in Theorem 1.1 is logarithmic, as for the min-entropy bound of [1]. A bound independent of k would be the conjecture for all DNF formulas, since a formula with m terms is read-m; on the examples we computed, including sunflower formulas, vertex stars of complete 2- and 3-uniform hypergraphs, graph formulas $\bigvee_{uv\in E}(x_u \wedge x_v)$ of random k-regular graphs with $k \le 10$ and random read-k formulas, H/I stays below 4.5 and does not grow with k. For unions of regular functions (Theorem 4.4) the logarithm enters in the same two ways: through the choice among at most k overlapping functions, and through the Cauchy–Schwarz step, whose factor k costs only $\log k$ because the rank tails and the level tails of the pieces decay exponentially. A second question is whether the biased form (6) holds for read-k DNF formulas and for regular functions under all product measures; together with [21] this would allow such functions as gates of read-once formulas, not only as inputs, and for threshold functions it would give the conjecture for all read-once threshold formulas. Every term on the left of (6) is nonnegative: computing $\mathbb{E}[g\phi_A]$ one coordinate at a time gives $P_\mu(A) = \sigma_A^2\, 4^{-|A|}\big(\mathbb{E}\,\partial_A g\big)^2$, where $\partial_A g(x) = \sum_{B\subseteq A}(-1)^{|A\setminus B|} g(x^{A\leftarrow 1_B})$ is the mixed difference in the coordinates of A and the expectation is over the other coordinates, so the left side of (6) equals $\sum_{A\neq\emptyset} P_\mu(A)\log\big(4^{|A|}/(\mathbb{E}\,\partial_A g)^2\big)$ and involves the biases of the coordinates in A only through the weight σ_A^2. The numerical evidence points to the constant of Lemma 5.5. For 60 threshold functions with at most 11 variables and 40 read-k DNF formulas with $k \le 3$ and at most 12 variables, with the product measure chosen by local search, the largest ratio of the two sides of (6), recomputed with 50-digit arithmetic, is below $2 + 4\log e$ and approaches it only at extreme biases, where the function behaves like AND_2. The same holds, in exact arithmetic, for every symmetric function of at most 8 variables and every symmetric threshold function of at most 30 variables under product measures with equal biases, at fifteen biases between 10^{-8} and $1 - 10^{-8}$. For symmetric threshold functions the ratio is bounded by Theorem 5.6 under equal biases and by Theorem 5.13 in general, with a larger constant. Its proof isolates what a general proof needs: after the rank step, the only obstacle is the normalisation $V\ln(1/V)$, which the chain rule splits into the marginal divergences of the conditional law μ_A, all of them repaid by the factors σ_A^2, and the total correlation of μ_A. For threshold functions under all product measures the total correlation is controlled by the tilt, which gives Proposition 5.10; when one input is pivotal with probability close to one, the other inputs are nearly deterministic and Theorem 5.13 follows by direct estimates. For general linear threshold functions the numerical bound $\alpha\,\mathrm{TC}(\mu_A) \le 0.31\, I_\mu$ of Section 6 is what a proof would have to establish, and the exponential tilt that gives it for symmetric thresholds does not: with unequal weights the Chernoff bound can exceed a rare event's probability by a factor that is not controlled by its influence. The inequality (6) itself held with the constant $2 + 4\log e$ of Lemma 5.5 for every linear threshold function that was computed, and a proof would extend Corollary 5.14 to read-once formulas over linear threshold gates of any fan-in.

A route that avoids the total correlation is to divide each set among its elements. For every Boolean function and every product measure,

$$\sum_{|A|\ge 2} P(A)\ln\frac{\sigma_A^2}{P(A)} = \sum_j \sigma_j^2\,\Psi(d_j), \qquad \Psi(d) = \sum_{B\neq\emptyset}\frac{\widetilde{d}(B)^2}{|B|+1}\ln\frac{\sigma_B^2}{\widetilde{d}(B)^2},$$

where d_j is the derivative in the j-th coordinate. For a function whose derivatives do not change sign, such as a linear threshold function, the sets of size one contribute at most $2J$, so a bound

$\Psi(d_j) \le C\mathrm{Var}(d_j)$ for every j gives (6) with constant $2(2+C)/\ln 2$. For a threshold gate d_j is the indicator of $\{N_{-j} = t-1\}$, whose coefficient on a set B of size b is $\prod_{i\in B}\sqrt{q_i(1-q_i)}$ times the b-th backward difference, at $t-1$, of the law of the number of ones outside B. For the conjunction of m inputs of bias q, the slice $N = m$, the ratio Ψ/Var tends to $\ln 4 + 2q\ln(1/q)/(1-q)$ as $m(1-q) \to \infty$, and every slice that was computed, with up to 256 inputs of equal bias or up to 8 inputs of unequal biases, satisfies $\Psi < (2+\ln 4)\mathrm{Var}$. For the conjunction this holds under every product measure: there $\widetilde{d}(B)^2 = \rho\prod_{i\in B}(1-q_i)\prod_{i\notin B} q_i$, and $q\ln(1/q) \le 1-q$ bounds the part of Ψ/Var coming from the inputs outside B by 2, so $2+\ln 4$ is the supremum over conjunctions. If $\Psi < (2+\ln 4)\mathrm{Var}$ held for all slices, (6) would follow for threshold gates with constant $2(4+\ln 4)/\ln 2 < 15.6$. When $\rho_j \ge \frac{1}{2}$, the anti-concentration bound used in the proof of Theorem 5.13 forces the other inputs to have σ^2-mass at most $930\mathrm{Var}(d_j)$, and Gibbs' inequality on each level then gives $\Psi(d_j) \le 7.04\mathrm{Var}(d_j)$; the open case is a slice far in the tail of the law of N_{-j}. For a linear threshold function the bound fails coordinate by coordinate: the derivative of a heavy input can be a threshold function of the light inputs, and then Ψ/Var grows like the logarithm of the number of light inputs, although the whole left side stays small. Other divisions of the sets among their elements do better: for $\mathbf{1}[m\,y_0 + y_1 + \cdots + y_m \ge m/2]$ with inputs of bias $\frac{1}{2}$ the equal division gives the heavy input the ratio 4.12 at $m = 512$, while giving each set to its lightest elements keeps every coordinate at 1.88, and the best division, found by linear programming, keeps every coordinate below 1.84 for every computed linear threshold function with at most 7 inputs. What is missing for linear threshold functions is an explicit division with this property.

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Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

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Einstein Institute of Mathematics, The Hebrew University of Jerusalem, Jerusalem, Israel
Email address: yitzchak.shmalo@gmail.com